ROSE5730 Autonomous Mobile Robots

Lecture 5: Localization



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Outline

Localization

3 Markov localization

2 Probabilistic localization

1 Belief and maps



Introduction



knowledge,
data base

mission
commands

Localization
Map Building

"position"
global map

Cognition
Path Planning

environment model
local map

path

Perception

Information
Extraction

raw data

Sensing

see-think-act

Path
Execution

actuator
commands

Acting

Motion Control

Real World
Environment

Localization

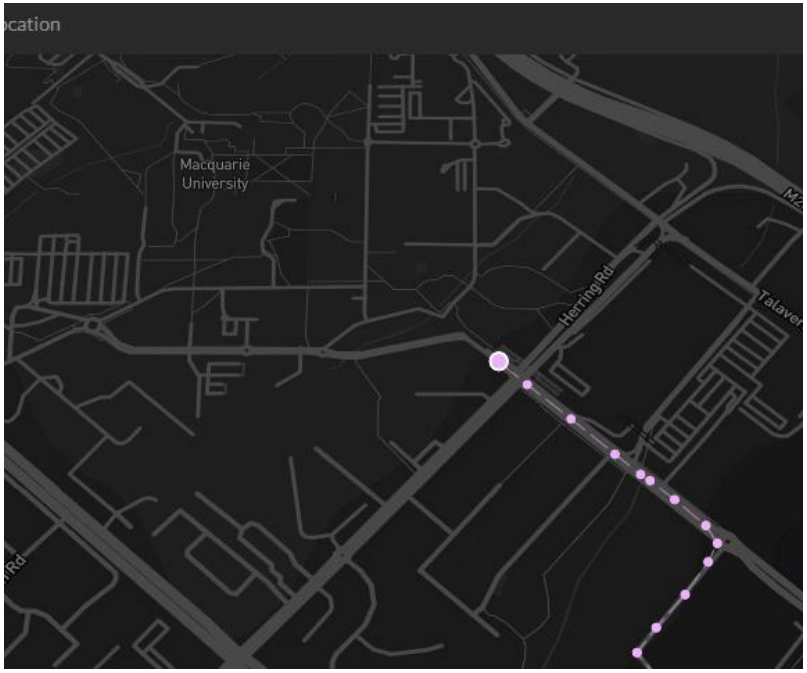
- Map-based localization
  - The robot estimates its position using perceived information and a map
  - The map might be known (Localization), or might be built in parallel (SLAM)
- Challenges
  - Measurements and the map are inherently **error prone**
  - The robot has to deal with **uncertain information** in prediction
  - Probabilistic map-base localization
- Approach
  - The robot estimates the belief state about its position through an SEE and ACT cycle
  - **Markov Localization**
  - **Kalman filter localization**

The **"Where am I?"** Problem



Three types of localization problems

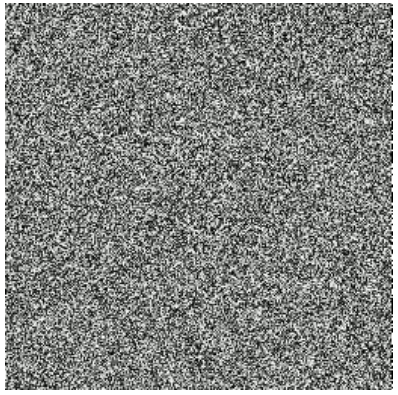
- Position Tracking (Local Localization)
  - Robot knows its initial position
  - Goal: Keep track of position as robot moves



- Global Localization
  - Robot does NOT know its initial position
  - Must determine position from scratch anywhere in environment

- Kidnapped Robot Problem
  - Robot thinks it knows where it is, but is wrong
  - Must detect and recover from localization failure

Three fundamental sources of difficulty







- Sensor Noise
  - Random errors in sensor measurements
- Sensor Aliasing (Perceptual aliasing)
  - Many different environmental states produce identical sensor readings
  - One-to-many mapping (one sensor reading → multiple possible locations)
- Effector Noise (Odometry Error)
  - Errors grow over time through movement

Representation decisions

- Two Fundamental Representation Decisions
  - Map Representation

    What aspects of the environment are contained in the map?

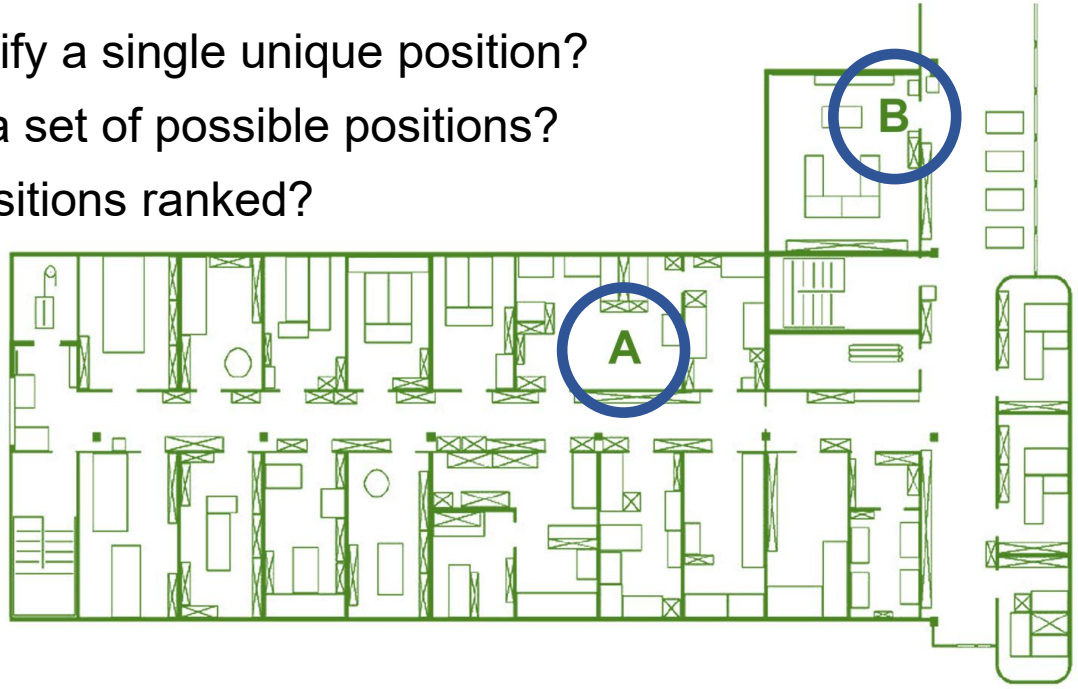
    At what level of fidelity does the map represent the environment?

  - Belief Representation

    Does the robot identify a single unique position?

    Or does it describe a set of possible positions?

    How are multiple positions ranked?

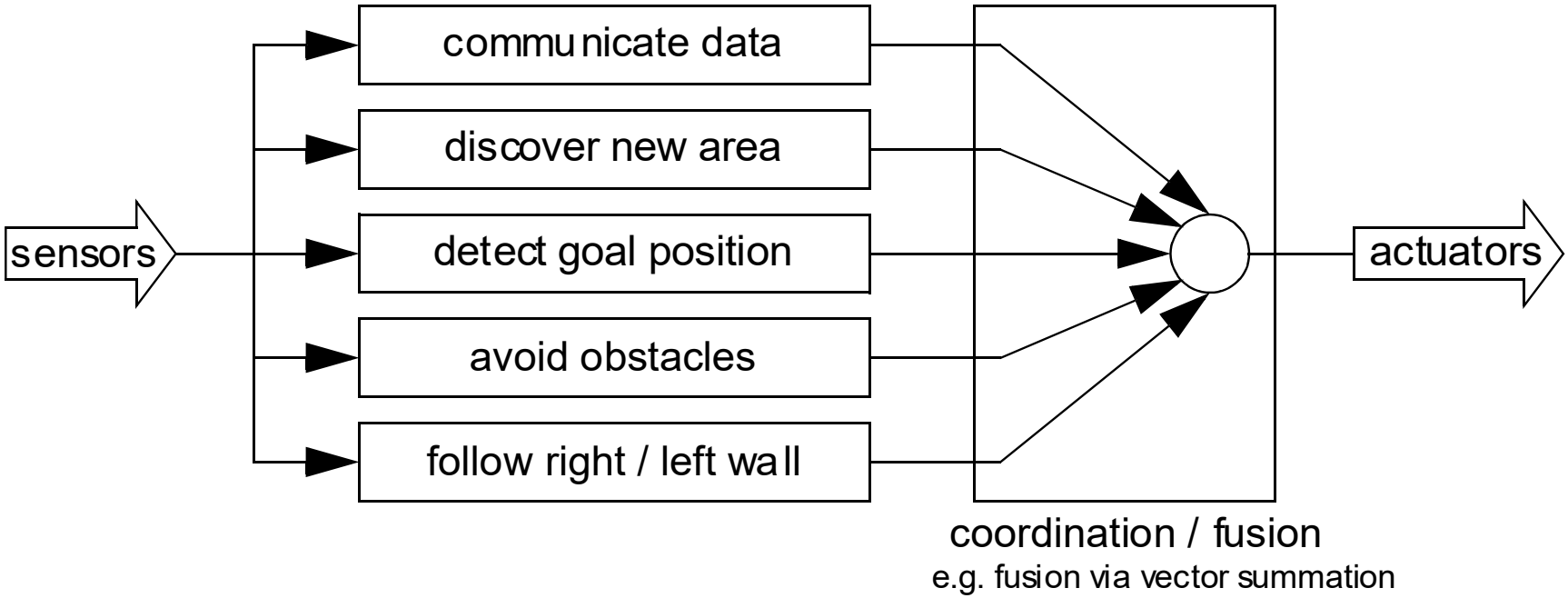


To localize or not to localize?

- Behavior-based Navigation
  - Avoids explicit localization - no geometric map
  - Uses sets of behaviors (e.g. wall-following, obstacle avoidance)
  - Goal detection through unique environmental cues
- Advantages
  - Quick implementation for specific environments
  - Simple architecture
- Disadvantages
  - Does not scale to new environments
  - Location-specific code
  - Behavior tuning is time-consuming
  - Adding new behaviors requires retuning all existing ones

To localize or not to localize?

- Behavior-Based Navigation

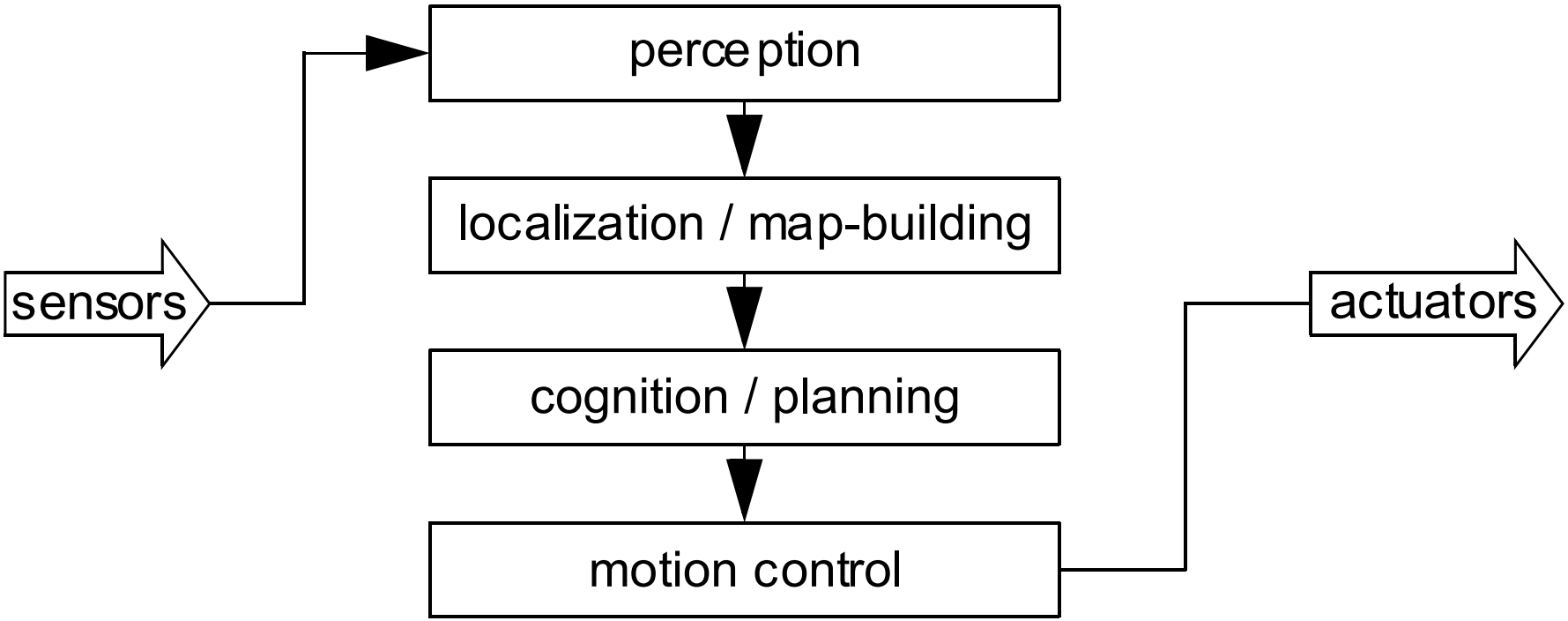


To localize or not to localize?

- Map-based Navigation
  - Explicit localization with reference to a map
  - Includes perception → localization → cognition → motion control pipeline
- Advantages
  - Transparent position belief for human operators
  - Map serves as communication medium between human and robot
  - Reusable architecture across environments
  - Map can be used by both robot and humans
- Disadvantages
  - Higher up-front development effort
  - Risk if internal model diverges from reality (i.e., if the map is wrong)

To localize or not to localize?

- Map-Based Navigation



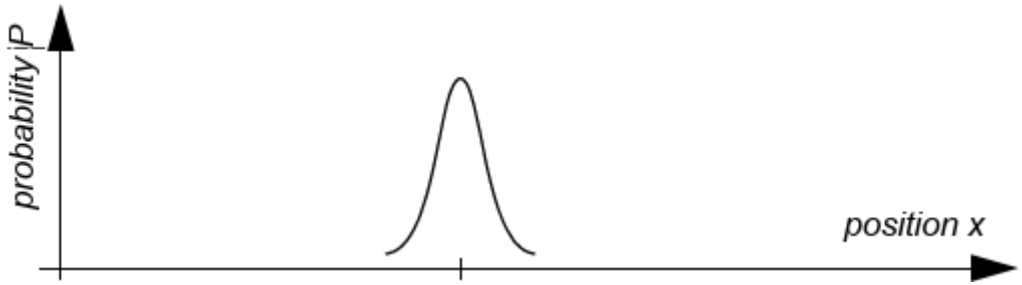
Belief representation

- **Pose**: the robot's state relative to a map (typically position and orientation, e.g., (x, y, θ) of a wheeled robot.
- **Belief**: the robot's estimate over possible poses, capturing both where it might be and how certain it is, formally a probability distribution over pose.
- **Belief representation**: the data structure/model used to encode that belief (e.g., a single pose with covariance, a probability grid, a mixture, or a set/region).
- **Why it matters**: the representation determines robustness to ambiguity, ability to recover when lost, memory/compute costs, and whether the robot can plan actions to reduce uncertainty.

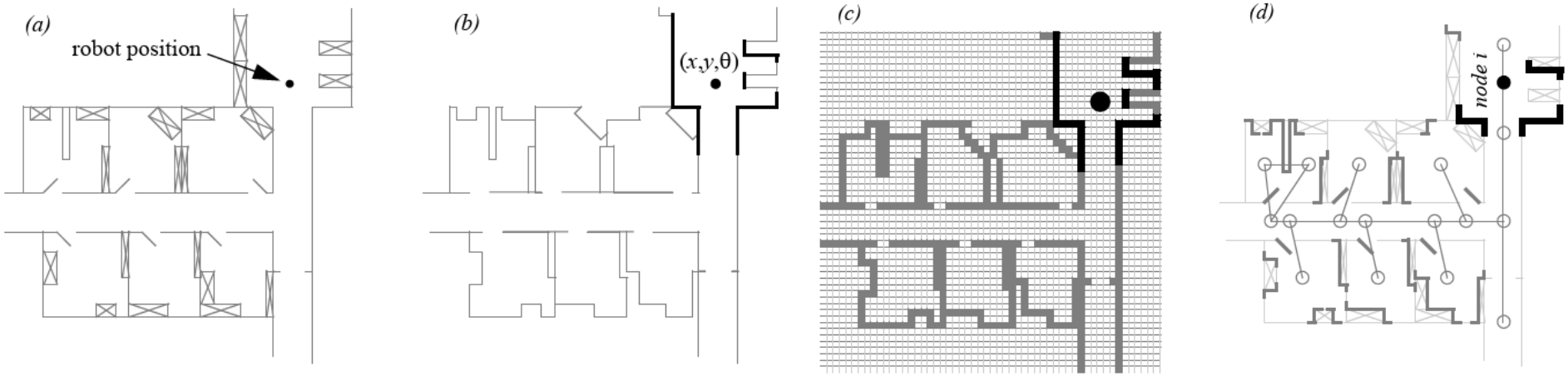
Single-hypothesis belief

- **Definition**: assume a single "best" pose at any time, i.e., **a unimodal belief**.
- Strengths
  - Simple & efficient: fast updates, compact state; easy to plug into control and planning.
  - Works well when initialized well and the environment is distinctive (low ambiguity).
- Limitations
  - Cannot represent ambiguity or multiple plausible locations; fragile under sensor aliasing.
  - Vulnerable to kidnapped-robot events (unexpected displacement) and global re-localization challenges.
- Typical realization includes **Kalman filter family**: maintains a mean pose and covariance, which is efficient but assumes unimodal Gaussian uncertainty.

Single-hypothesis belief



Continuous map with single-hypothesis belief, e.g., single Gaussian centered at a single continuous value.



Examples of single hypotheses of position using different map representations: (a) real map with walls, doors and furniture; (b) line-based map; (c) occupancy grid-based map; (d) topological map using line features.

Multiple-hypothesis belief

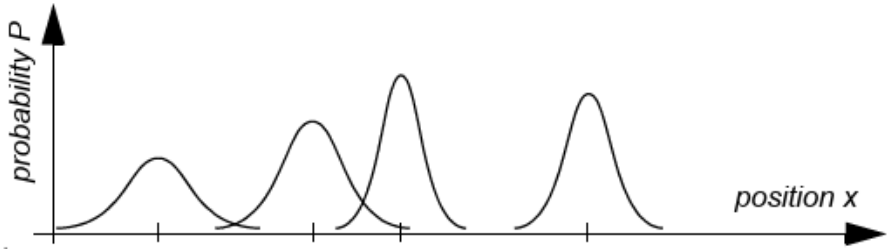
- **Definition**: maintain many possible poses (a set/region or probability distribution over pose space) with likelihoods and captures multi-modal uncertainty.
- Strengths:
  - Robust to ambiguity and loss of localization; enables global localization and recovery.
  - Supports information-seeking actions (plan motions to disambiguate).
- Limitations:
  - Heavier compute and memory requirement;
  - number of hypotheses can grow with map size or ambiguity.
- Typical realizations：

  Discrete probability grids including **Markov and histogram localization**

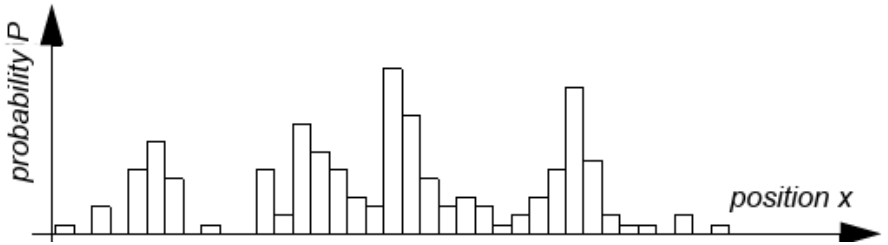
  Gaussian mixtures or geometric feasible-pose regions to approximate multi-modality.

  Sample-based model as a non-parametric multi-hypothesis encoding

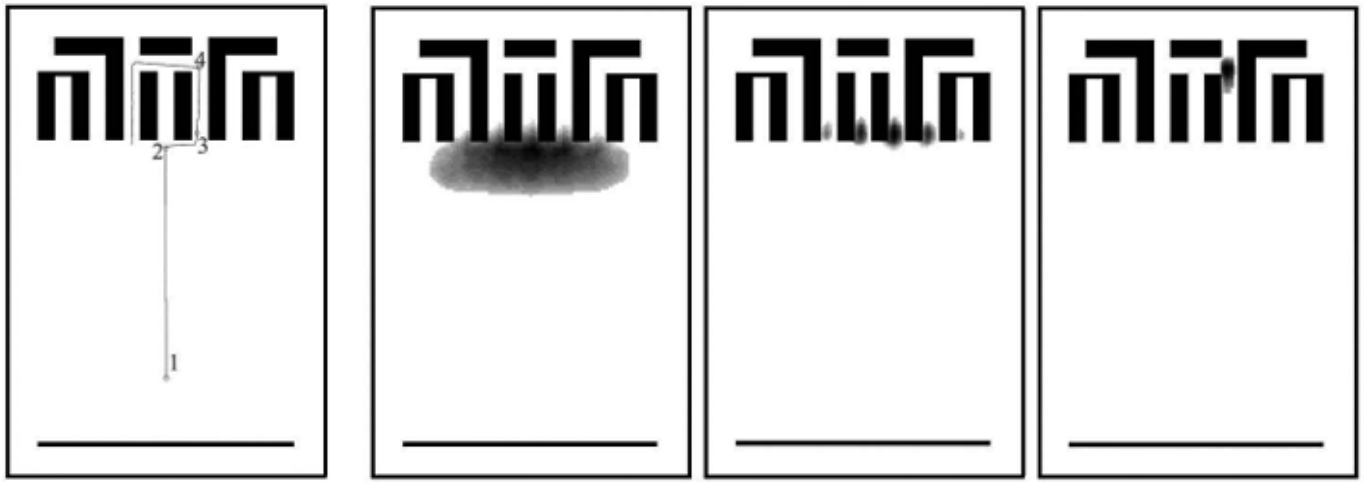
Multiple-hypothesis belief



Continuous map with multiple gaussians.



Discretized grid map with probability values.



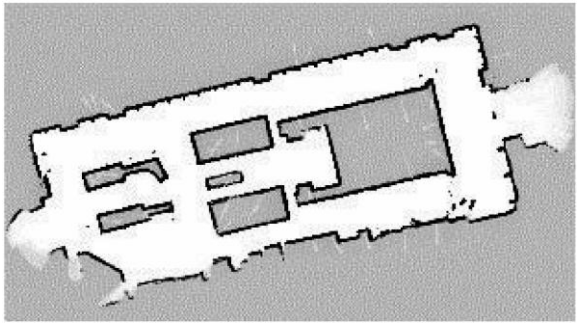
Path of the robot

Belief states at positions 2, 3, and 4

Example of multiple-hypothesis tracking. The belief state that is largely distributed becomes very certain after moving to position 4. Note that darker coloring represents higher probability.

Map representation

- **Purpose**: A map encodes the environment so the robot can localize, plan, and navigate. The chosen representation strongly affects accuracy, runtime/memory cost, and how well sensor data can be integrated.





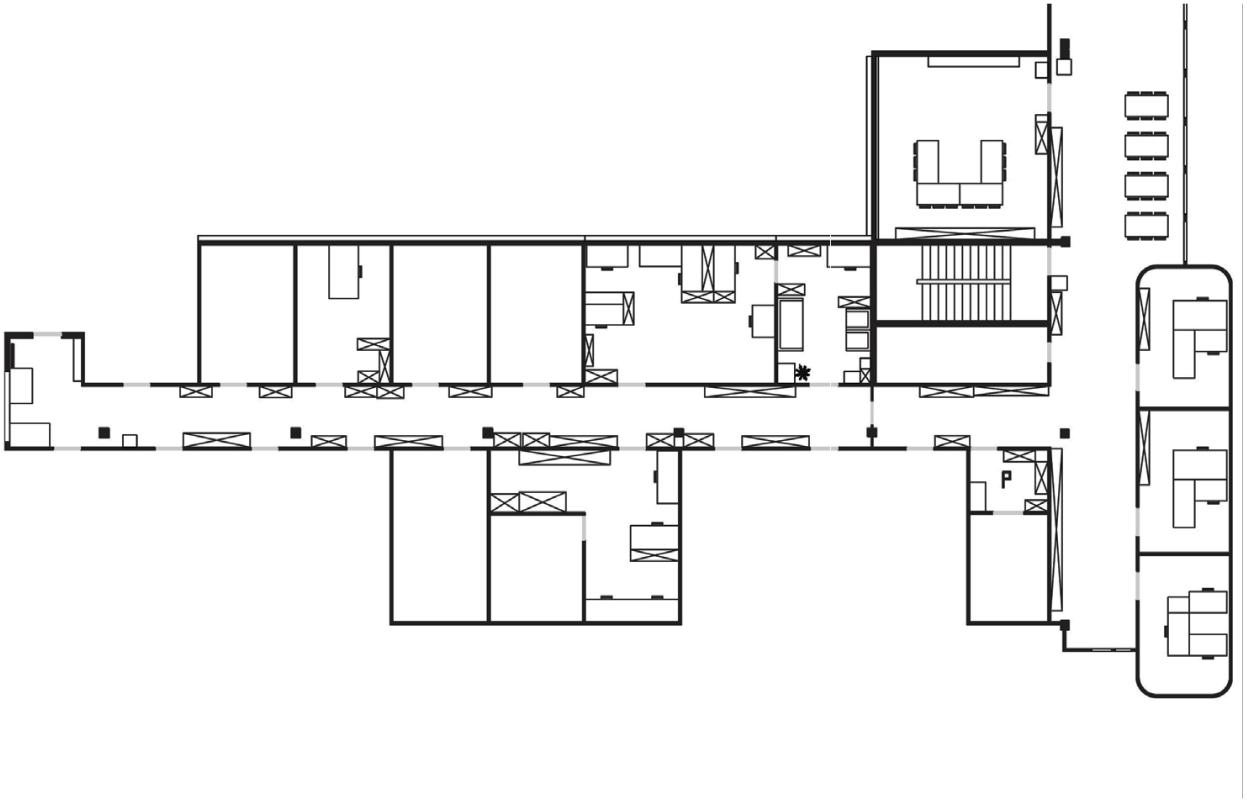
- Three Fundamental Relationships for Map Design
  - **Precision matching**: Map precision must match the precision needed to achieve robot goals
  - **Sensor compatibility**: Map precision and feature types must match what robot sensors can detect
  - **Computational complexity**: Map representation complexity directly impacts localization and navigation computation

Continuous representations

- Continuous Representations
  - Exact decomposition of environment in continuous space
  - Commonly used in 2D representations only (higher dimensions cause computational explosion)
  - Often uses closed-world assumption: only specified objects exist; empty areas truly have no objects
- Polygon-based
  - Architecture map
- Line-based
  - Representation with set of finite or infinite lines

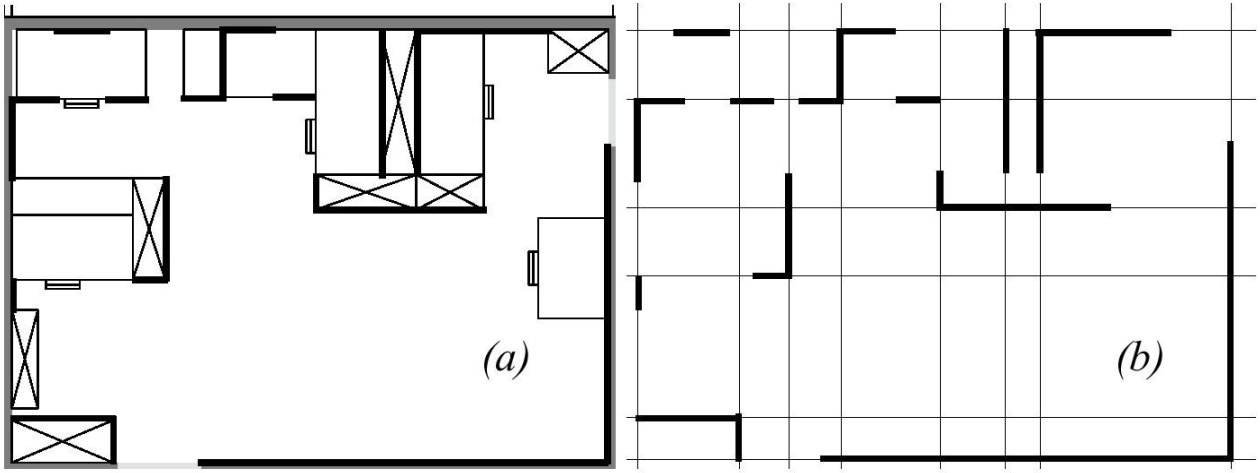
Continuous representations

- Polygon-based
  - Architecture map



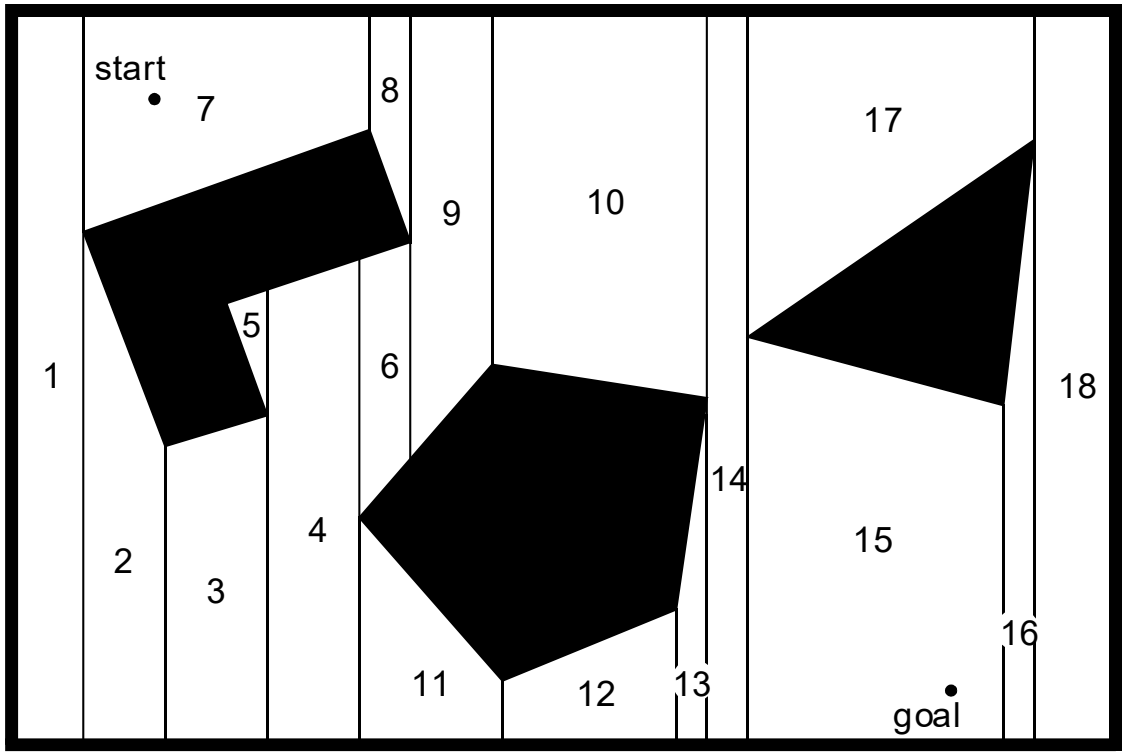
Continuous representations

- Line-based
  - Representation with set of finite or infinite lines



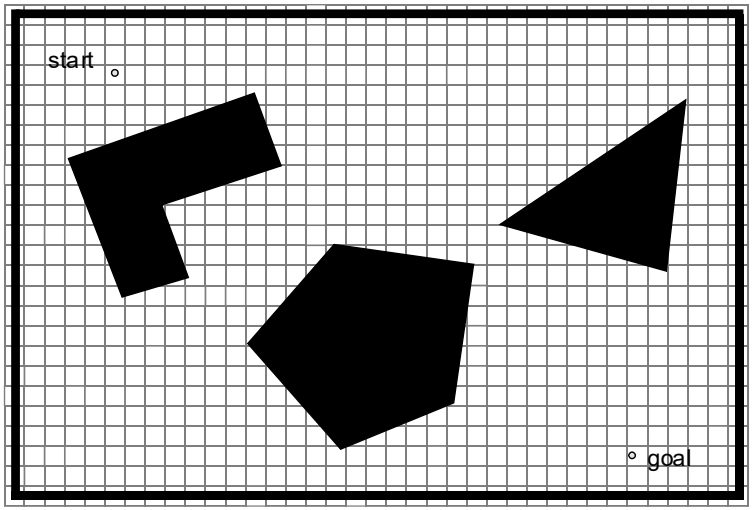
Decomposition (discrete) strategies

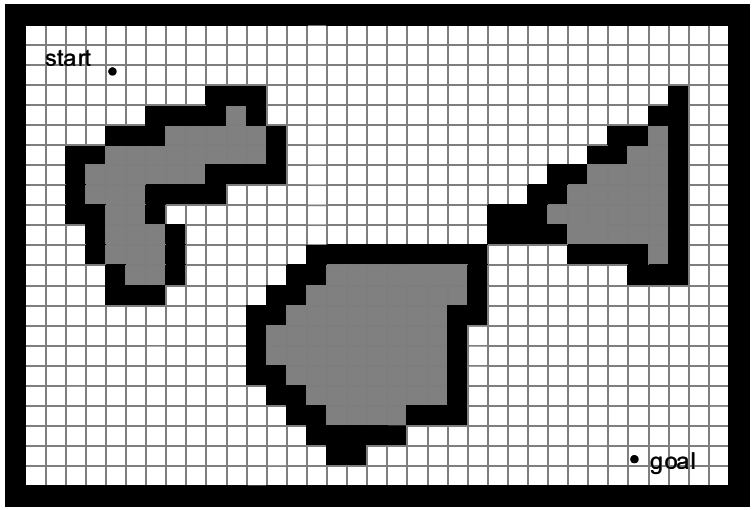
- Decomposition Strategies: Discretize space into cells or regions, and attach simple attributes
- Exact cell decomposition - Polygons



Decomposition (discrete) strategies

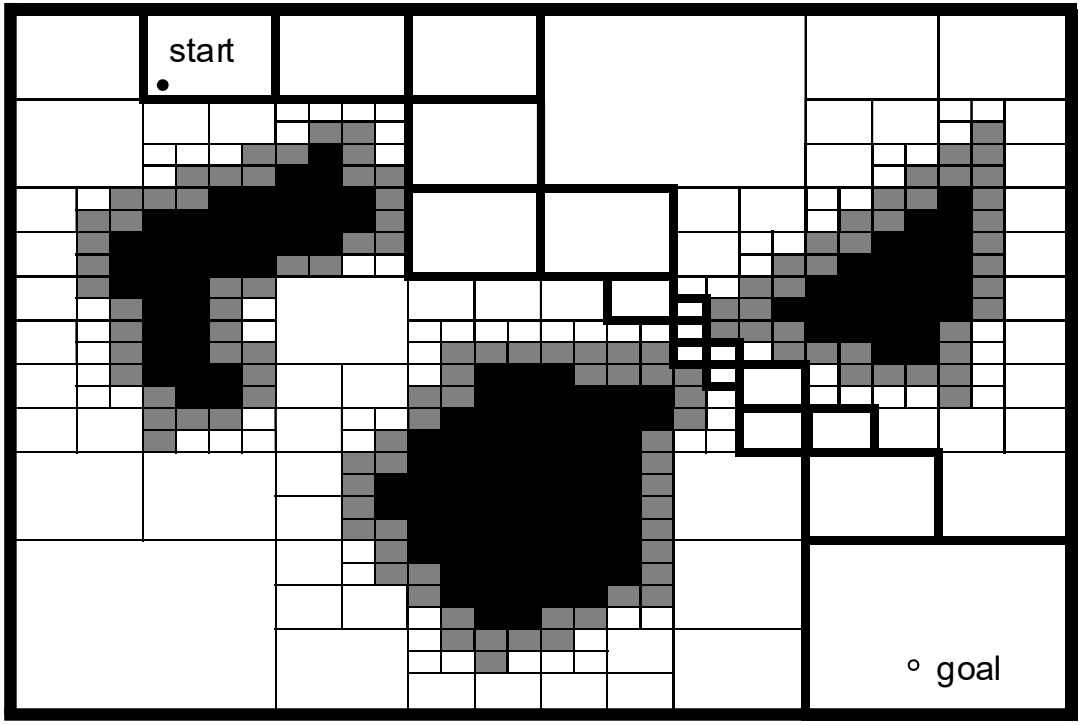
- Approximate cell decomposition: Partition into a uniform grid (e.g., 5 cm $\times$ 5 cm). Each cell stores the probability of occupancy, updated from range data.





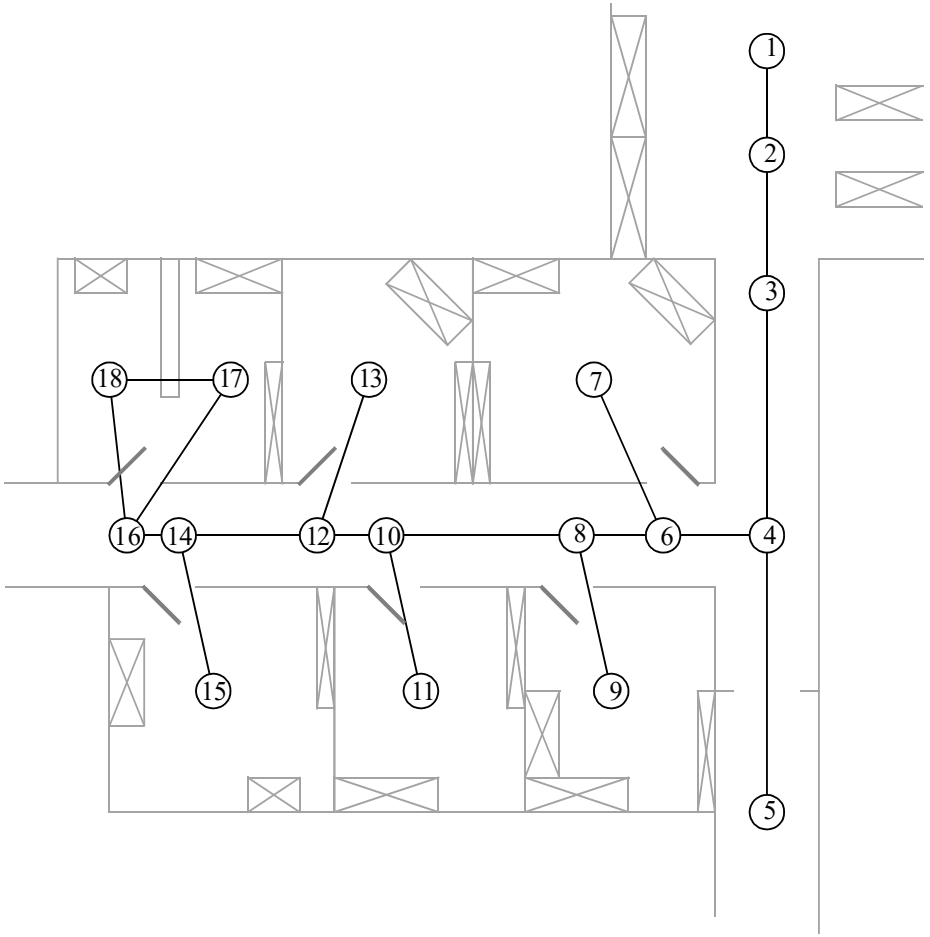
Decomposition (discrete) strategies

- Adaptive cell decomposition: Use variable cell sizes (e.g., quadtrees): fine cells where detail is needed, coarse elsewhere.



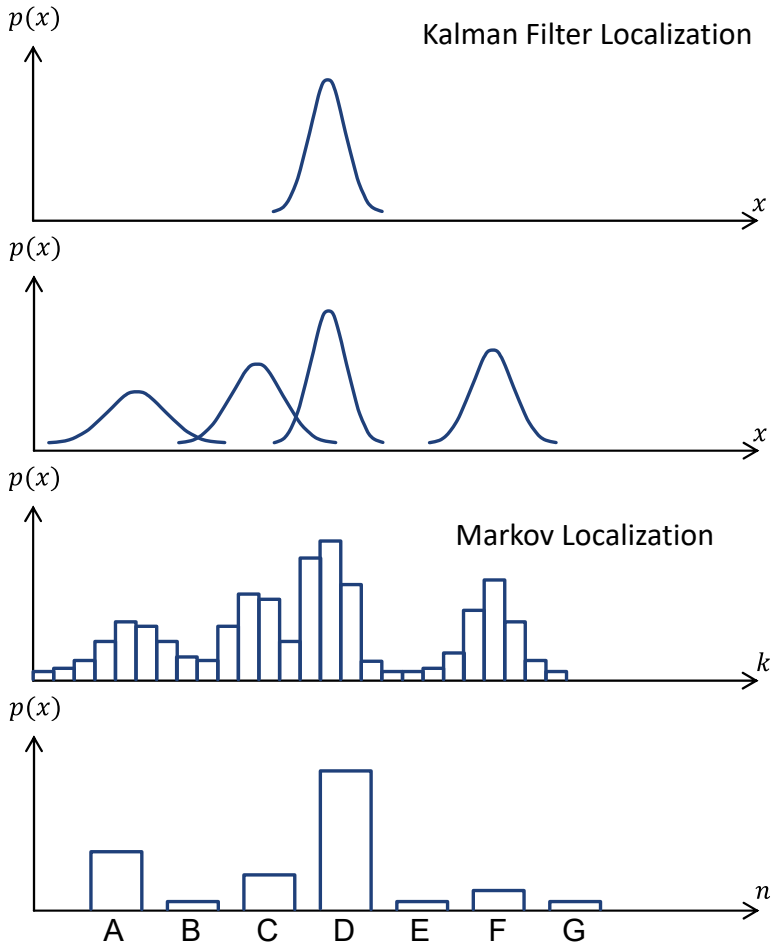
Topological map representation

- Topological map
  - Represents the environment as a graph with nodes and edges
  - Nodes correspond to spaces
  - Edge correspond to physical connections between nodes
  - Lack scale and distances, but topological relationships (e.g., left, right, etc.) are maintained

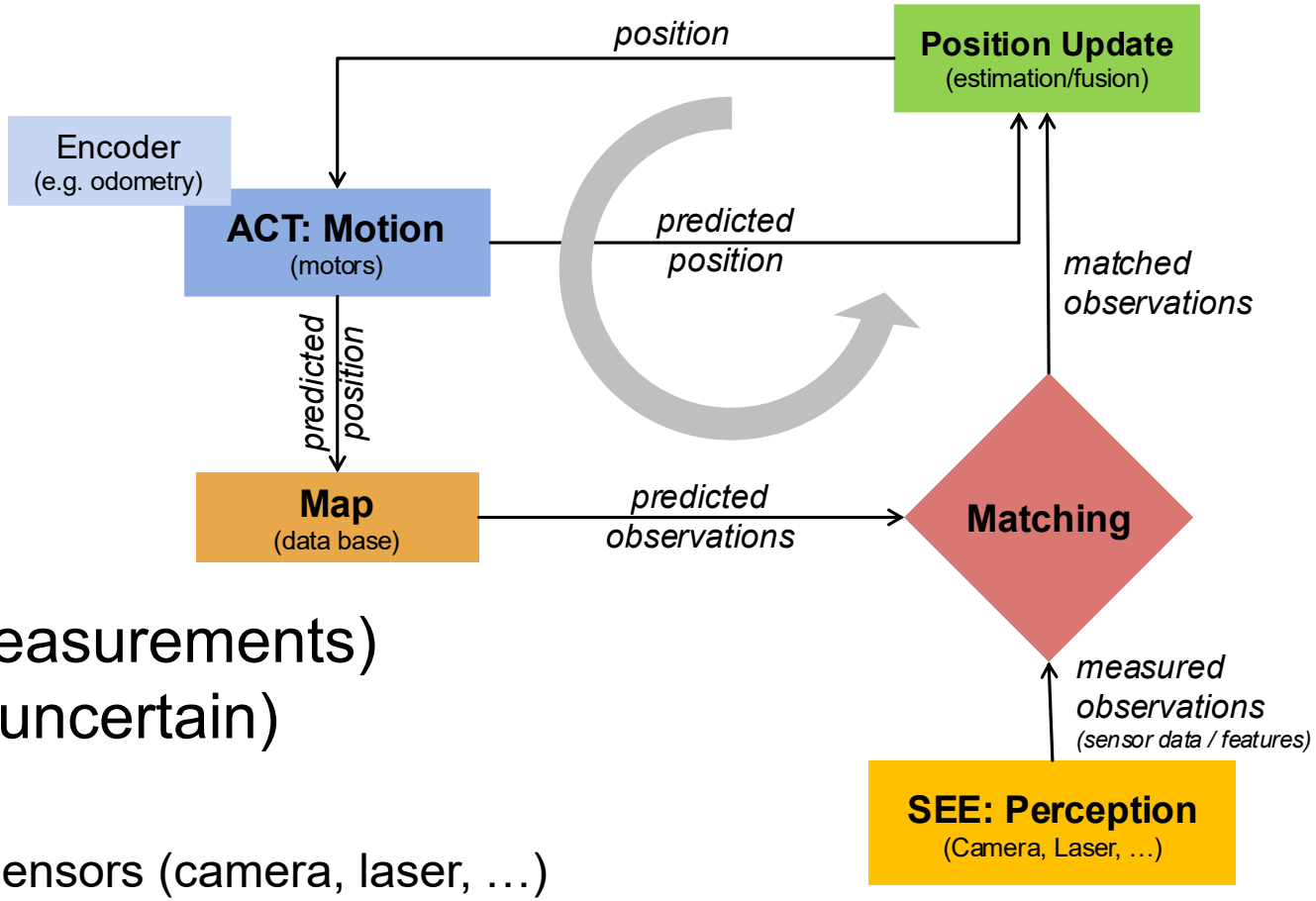


Recap: Belief representation

- Continuous map with single hypothesis probability distribution p(x)
- Continuous map with multiple hypotheses probability distribution p(x)
- Discretized metric map (grid k) with probability distribution p(k)
- Discretized topological map (nodes n) with probability distribution p(n)



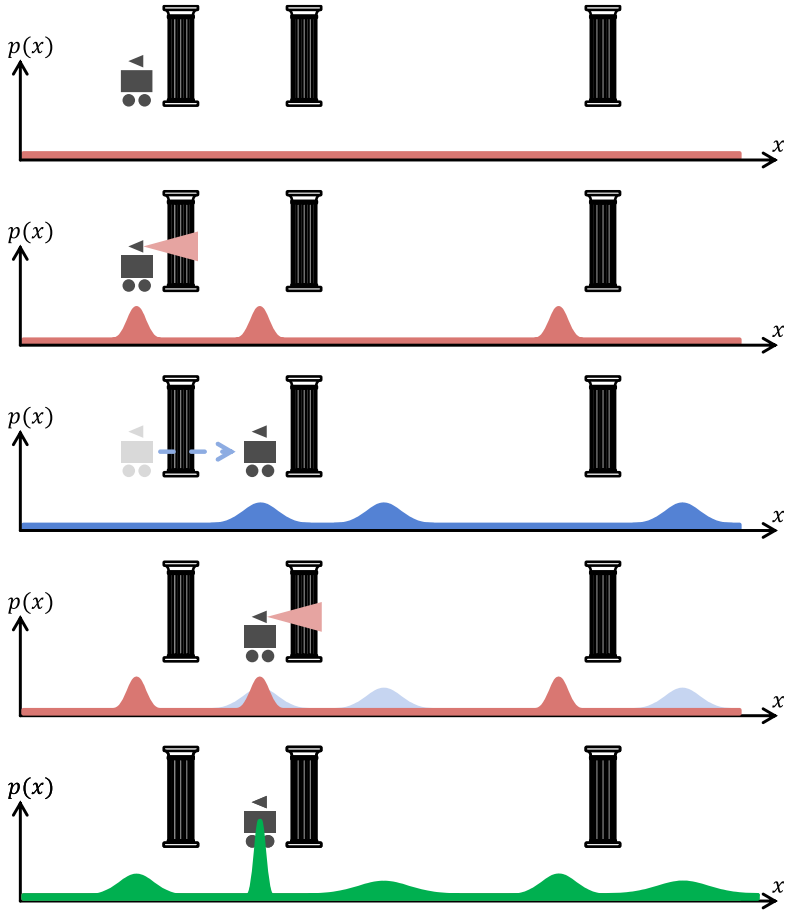
Map-based localization
The estimation cycle (ACT-SEE)



- Information (measurements) is error prone (uncertain)
  - Odometry
  - Exteroceptive sensors (camera, laser, …)
  - Map
- → Probabilistic map-based localization

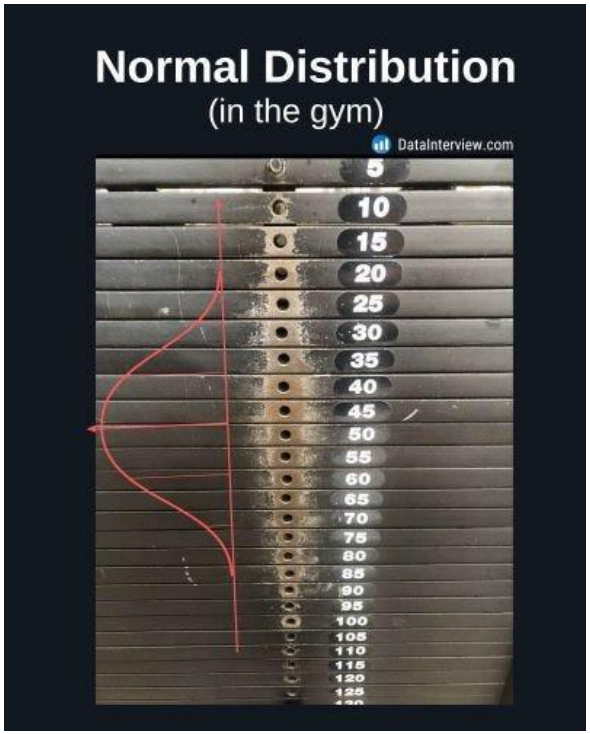
ACT-SEE cycle for localization

- Robot is placed somewhere unknown
- SEE: The robot queries its sensors
  - → finds itself next to a pillar
- ACT: Robot moves one unit forward
  - motion estimated by wheel encoders
  - accumulation of uncertainty
- SEE: The robot queries its sensors again
  - → finds itself next to a pillar
- Belief update (information fusion)



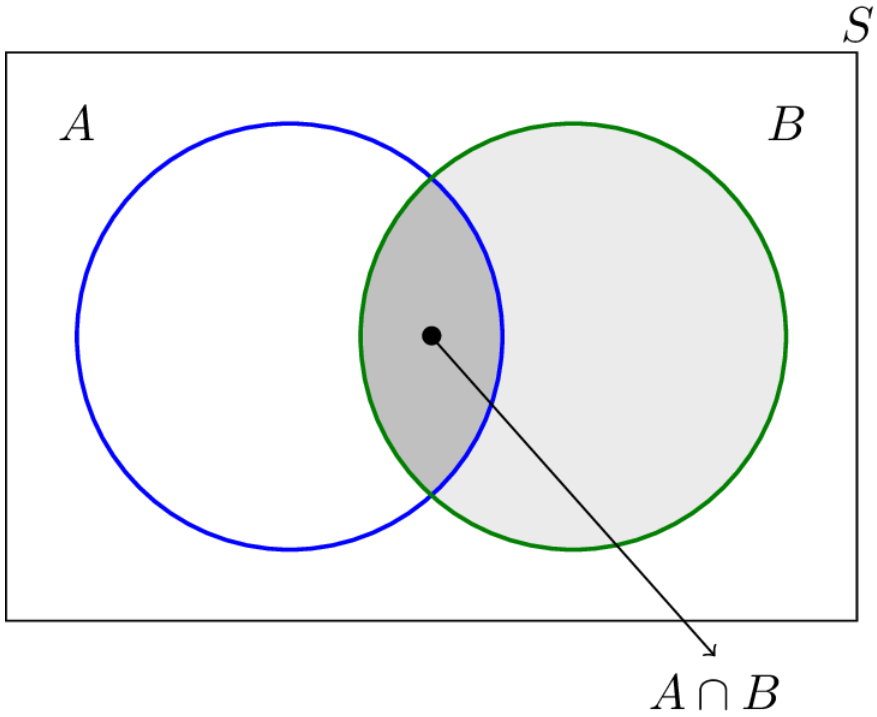
Recap probability theory

- Probability Revision
  - $p(X = x)$
- The Probability Density Functions (PDF)
  - The relative likelihood for a random variable to take on a given value
- Gaussian Distribution
  - $p(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$
  - $p(x) = N(\mu, \sigma^2)$
  - where μ and σ^2 specify the mean and variance



Recap probability theory

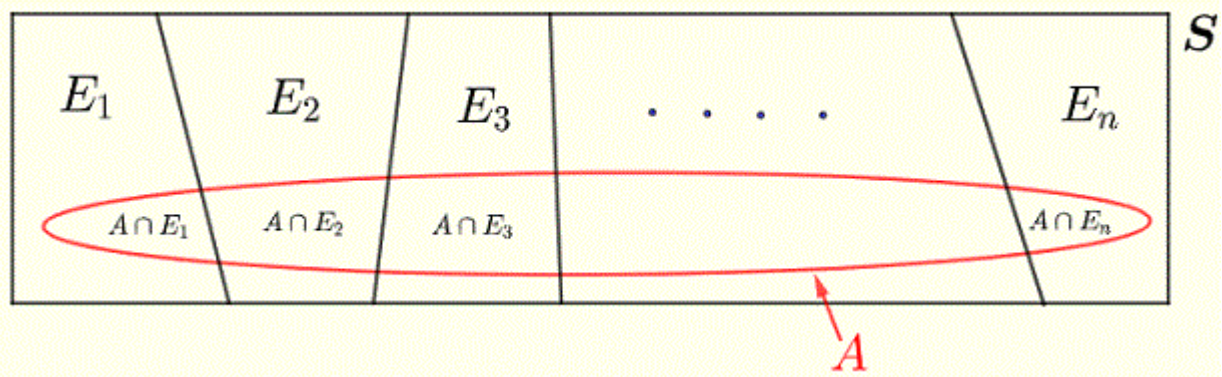
- Joint Distribution
  - $p(x, y) = p(x)p(y)$
    (X, Y are independent)
- Conditional Probability
  - $p(x|y) = \frac{p(x,y)}{p(y)}$
  - If X and Y are independent:
    $p(x|y) = \frac{p(x)p(y)}{p(y)} = p(x)$



$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Recap probability theory

- Theorem of total probability
  - $p(x) = \sum_y p(x|y)p(y)$
  - $p(x) = \int_y p(x|y)p(y)dy$



Example: Robot Sensor Fusion

A mobile robot is moving down a hallway and wants to detect whether the next area contains a door. It uses a laser scanner to classify each segment of wall as either:

- D = “There is a door”
- W = “There is no door (just a wall)”

Probability of encountering a door in this hallway: P(D) = 0.2 Probability of encountering a solid wall: P(W) = 0.8

The robot's laser classifier has known sensor characteristics: If there is a door, the sensor correctly detects it 90% of the time:

P(sensor = “door detected” | D) = 0.9

If there is no door, the sensor still mistakenly reports a door 10% of the time (false positive):

P(sensor = “door detected” | W) = 0.1

Question: Compute the overall probability that the sensor will report “door detected”, taking into account both situations (door or wall).

Recap probability theory

- The Bayes Rule

$$p(x|y) = \frac{p(y|x)p(x)}{p(y)}$$

$$p(x|y) = \eta p(y|x)p(x) \qquad \eta = p(y)^{-1}$$



Example: CU shuttle bus in the rain



At CUHK, students often choose between:

- Taking the shuttle bus
- Walking across campus

From past observation: P(Rain) = 0.3, P(Sun) = 0.7

Students behave differently depending on the weather:

- If rainy, 90% of students take the shuttle bus: P(Bus|Rain)=0.9
- If sunny, only 40% take the shuttle bus: P(Bus|Sun)=0.4



Question: If we find a random CUHK student taking the shuttle bus, what is the probability that it is raining?

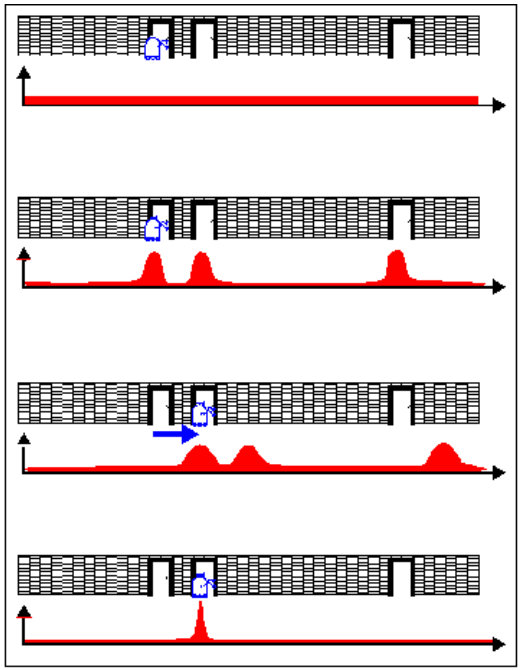
Markov localization

- Discretized pose representation x_t → grid map
- Tracks the robot's belief state $bel(x_t)$ using an arbitrary PDF to represent the position
- Markov assumption: The output of the estimation process is a function x_t, only of the robot's previous state x_{t-1}, and its most recent actions (odometry) u_t and perception z_t.

$$p(x_t|x_{t-1}, u_t, z_t)$$

Markov localization

- **ACT** | Probabilistic estimation of
  - the robot's new belief state $\overline{bel}(x_t)$, based on the previous location $bel(x_{t-1})$, and the probabilistic motion $p(x_t|u_t, x_{t-1})$, with the action u_t (control input)



- Apply **Theorem of Total Probability**
  - $\overline{bel}(x_t) = \int p(x_t|u_t, x_{t-1}) bel(x_{t-1}) dx_{t-1}$, for continuous probabilities
  - $\overline{bel}(x_t) = \sum p(x_t|u_t, x_{t-1}) bel(x_{t-1})$, for discrete probabilities

- = Convolution!!

Markov Localization

- **SEE** | Probabilistic estimation of
  - the robot's new belief state $bel(x_t)$,
    as a function of its measurement data z_t,
    and its former belief state $\overline{bel}(x_t)$

- Apply **Bayes Rule**
  - $bel(x_t) = \eta p(\boldsymbol{z_t}|x_t, M)\overline{bel}(x_t)$
  - $p(z_t|x_t, M)$ is the probability of observing the measurement data $\boldsymbol{z_t}$ given the map M and position x_t
  - $\eta = p(y)^{-1}$ is the normalization factor so that $\sum p = 1$

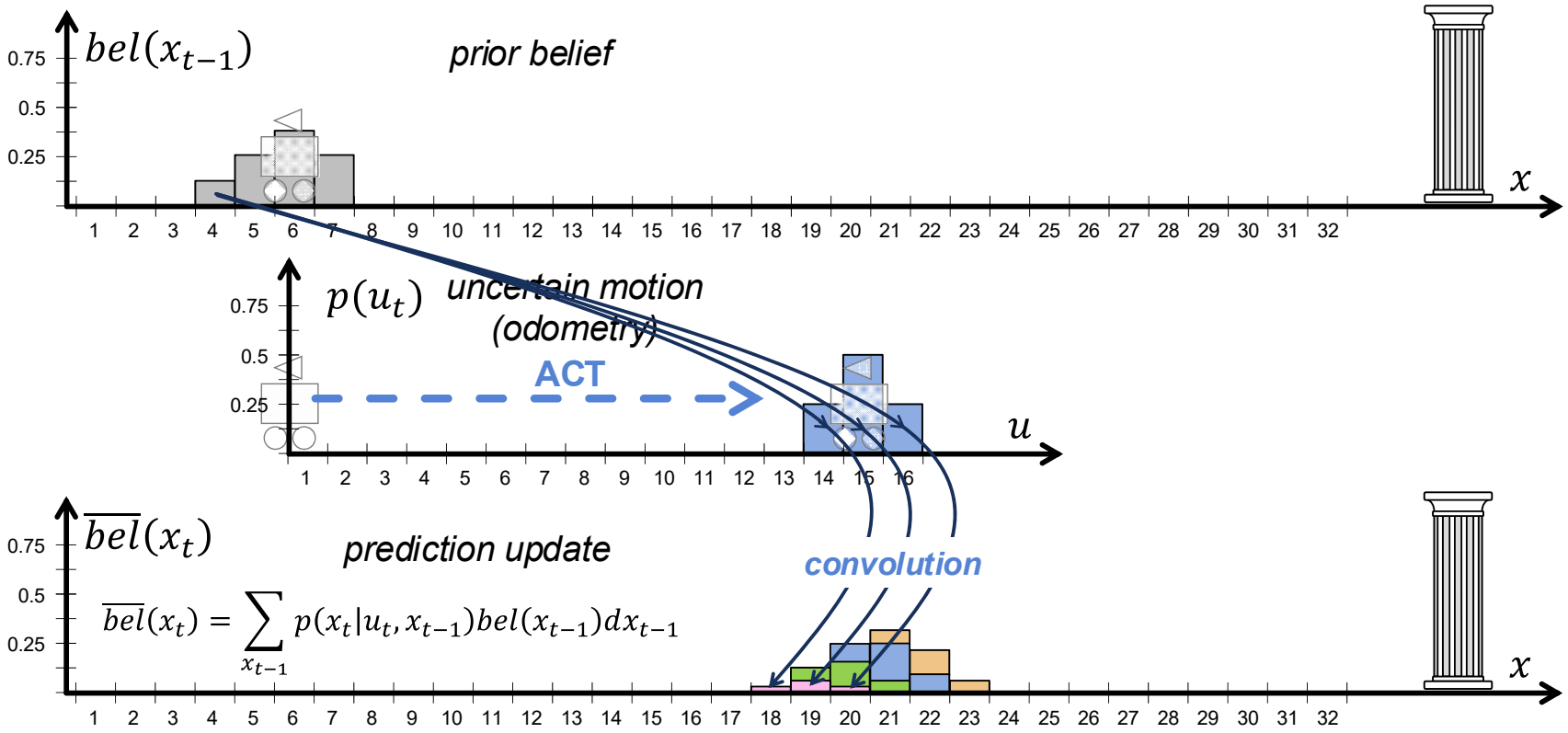
Markov localization

- Basic algorithms
  - For all x_t do
    - $\overline{bel}(x_t) = \sum_{x_{t-1}} p(x_t|u_t, x_{t-1}) bel(x_{t-1})$ (prediction update)
    - $bel(x_t) = \eta p(z_t|x_t, M) \overline{bel}(x_t)$ (measurement update)
  - endfor
  - return $bel(x_t)$

- **Markov assumption**: The output of the estimation process is a function x_t, only of the robot's previous state x_{t-1}, and its most recent actions (odometry) u_t and perception z_t.

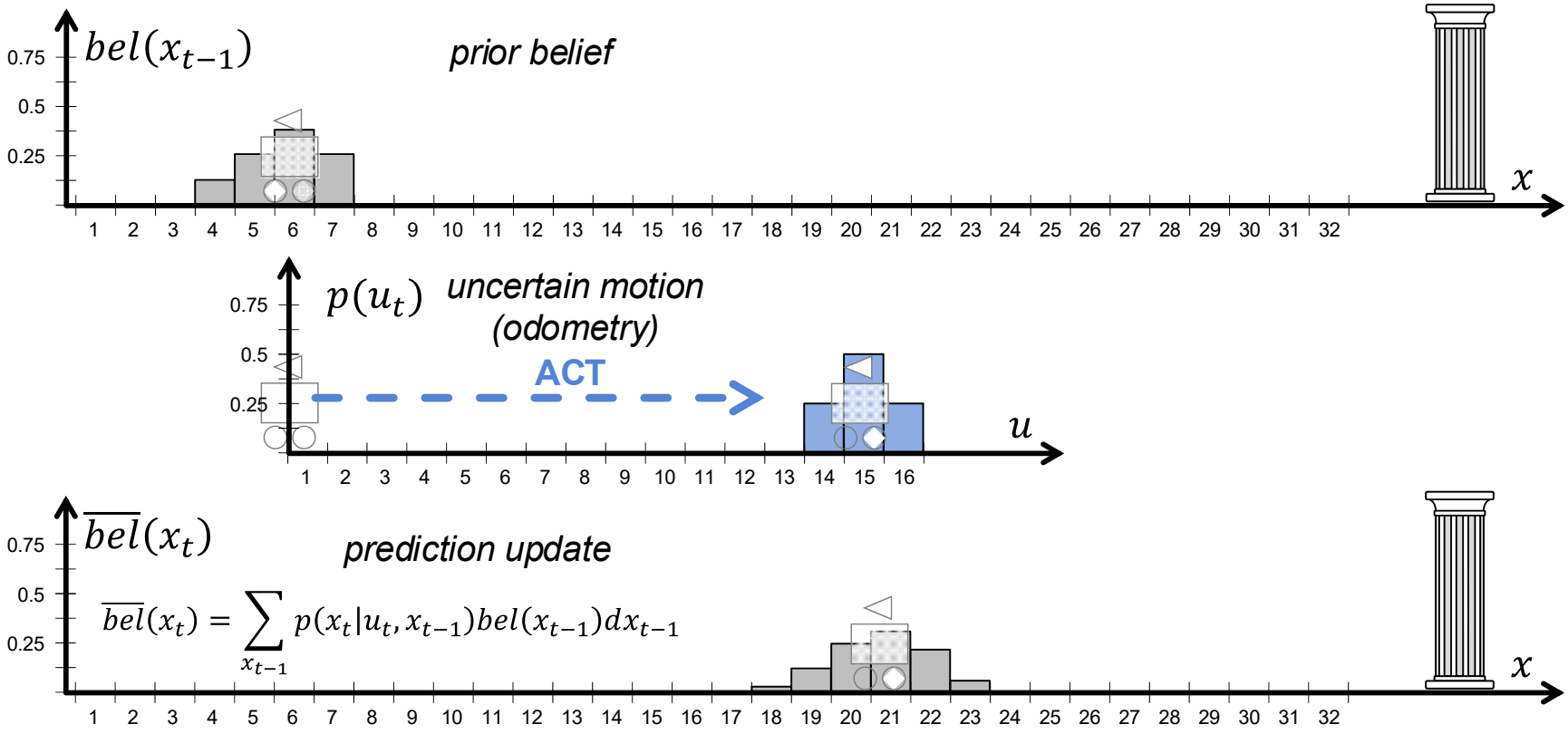
Markov localization

- ACT | using motion model and its uncertainties



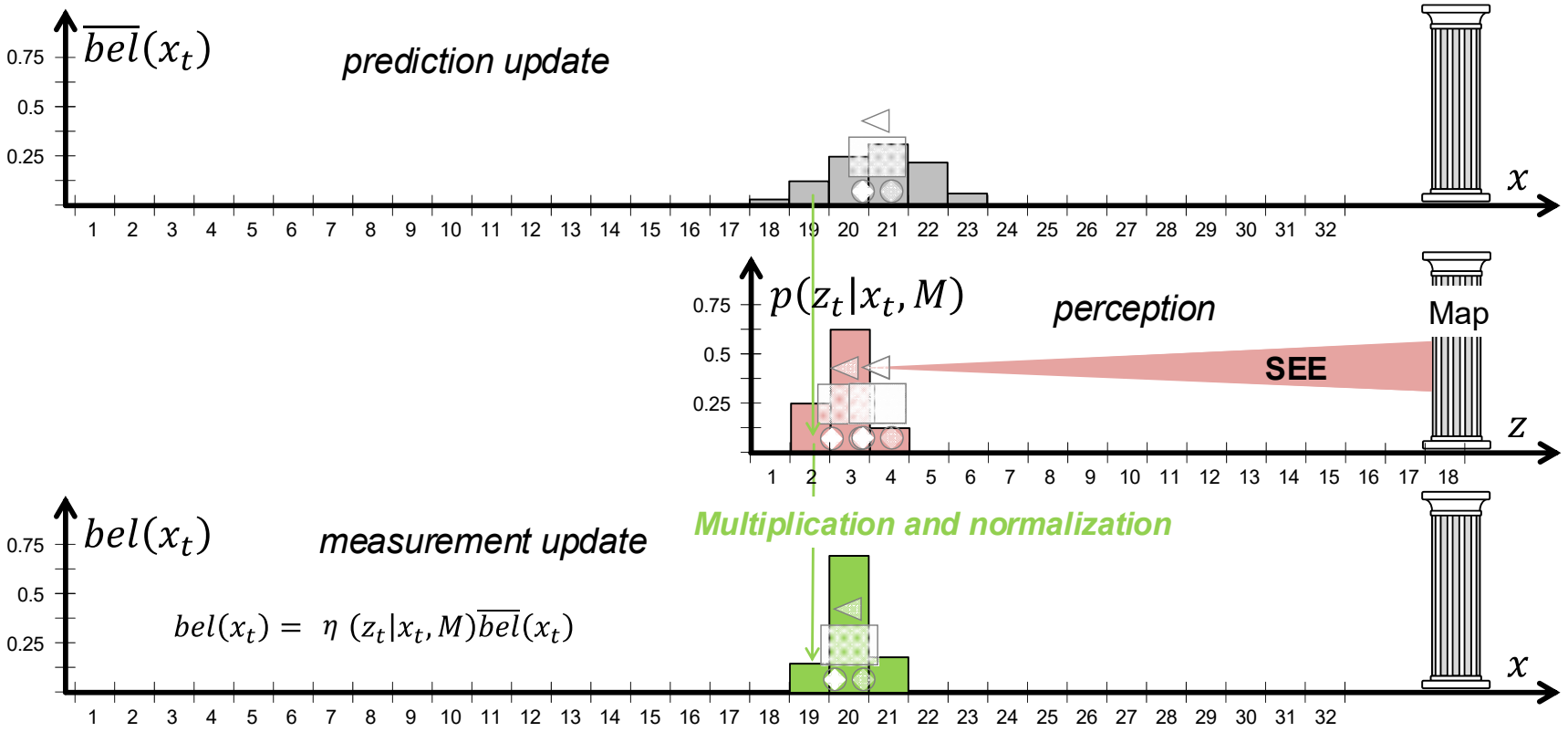
Markov localization

- ACT | using motion model and its uncertainties



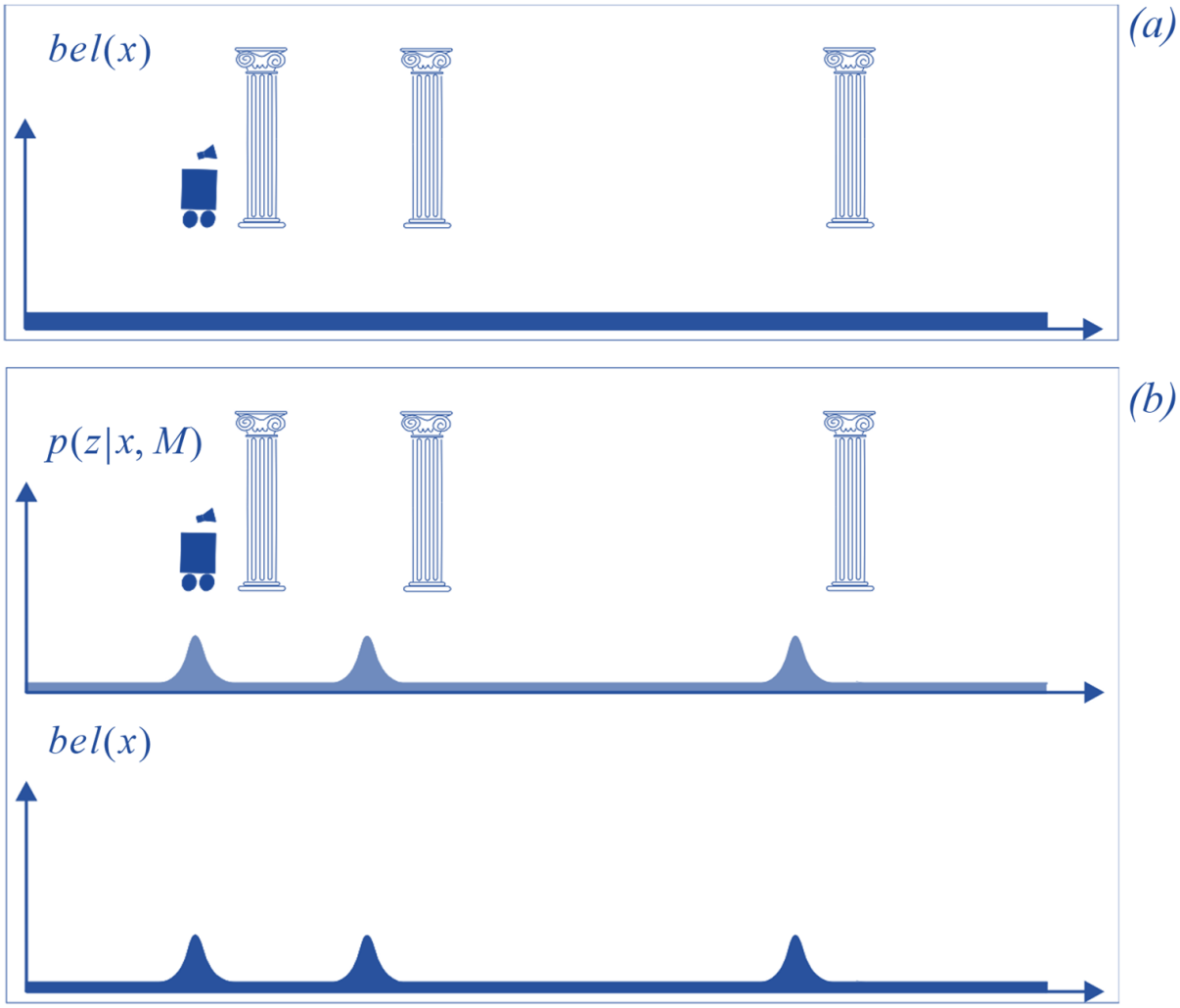
Markov localization

- SEE | using motion model and its uncertainties



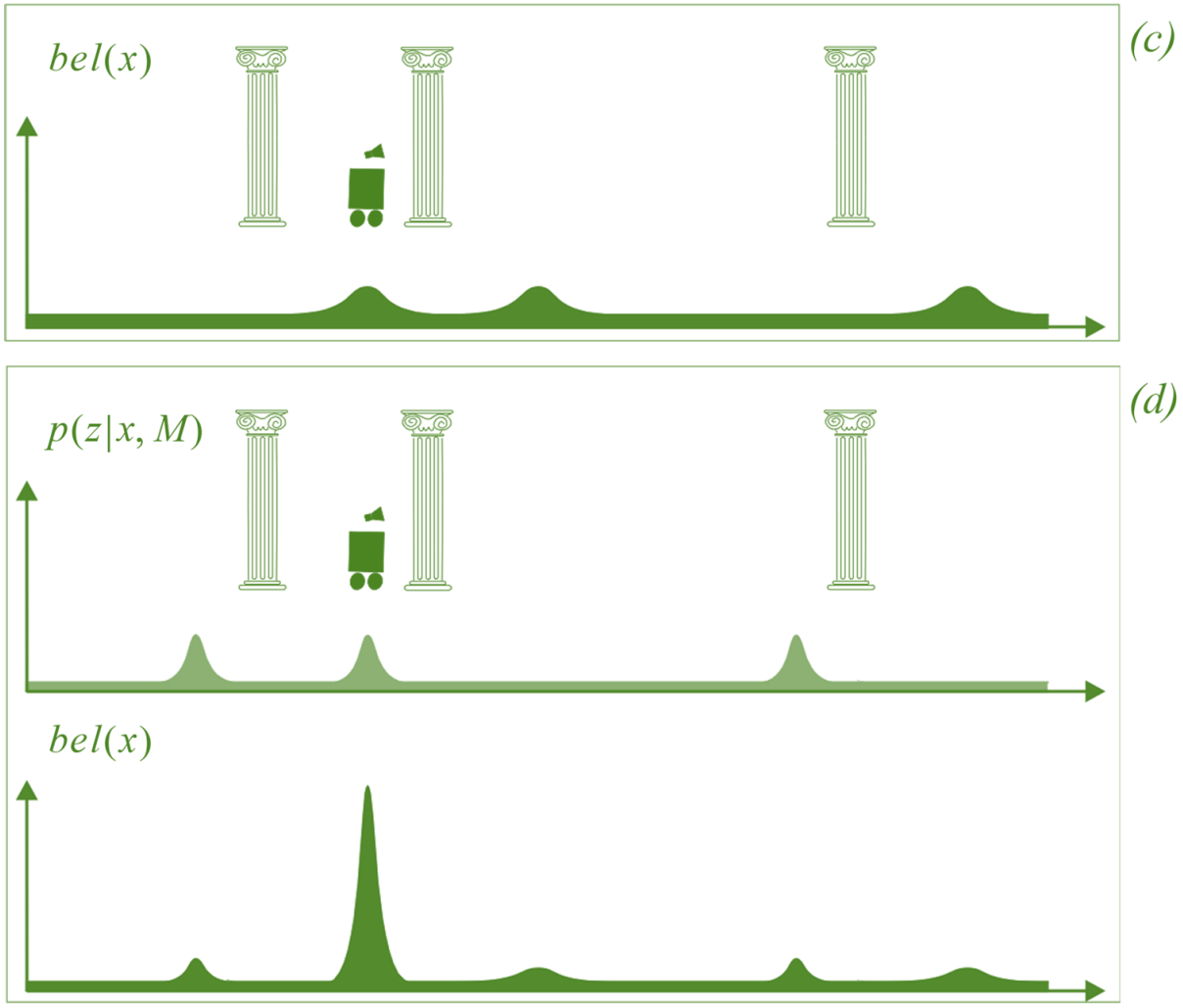
Markov localization

- **ACT |** using motion model and its uncertainties



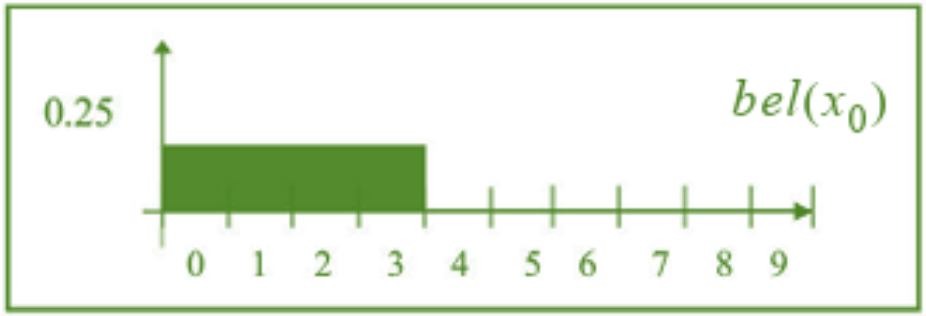
Markov localization

- **SEE** | using motion model and its uncertainties



Markov localization: 1D case

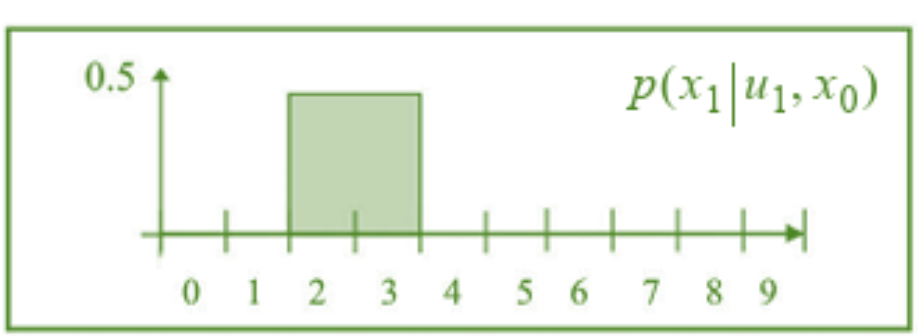
- Assume the environment is one-dimensional for illustration.
  - The environment (one segment on an axis) is tessellated into ten equally spaced cells.
- Suppose that the robot's initial belief $bel(x_0)$ is a uniform distribution from 0 to 3.
  - Observe that all the elements were normalized so that their sum is 1
  - $p(x_0 = 0) = 0.25, p(x_0 = 1) = 0.25, p(x_0 = 2) = 0.25, p(x_0 = 3) = 0.25$



(a)

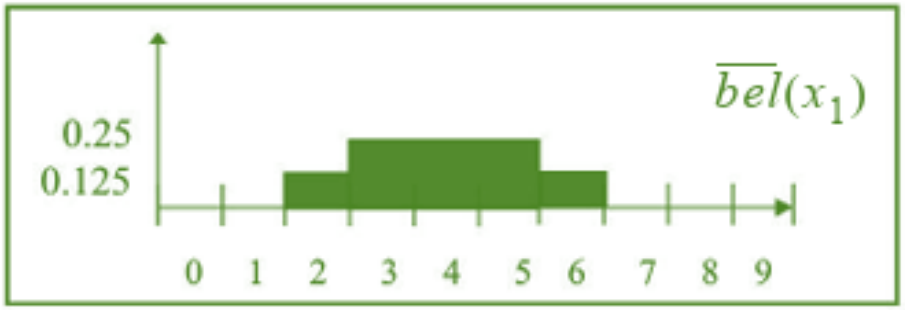
1D case: Prediction update

- In this phase, the robot moves and updates its belief using the motion model of the control input.
  - The probabilistic motion model (i.e. the odometric error model) is needed for computing $\overline{bel}(x_1)$.
  - The probabilistic motion model of the odometry $p(x_1|u_1, x_0)$ is given as: between time $t = 0$ and time $t = 1$, the robot may have moved either two or three units to the right. Both movements have the same probability to occur.



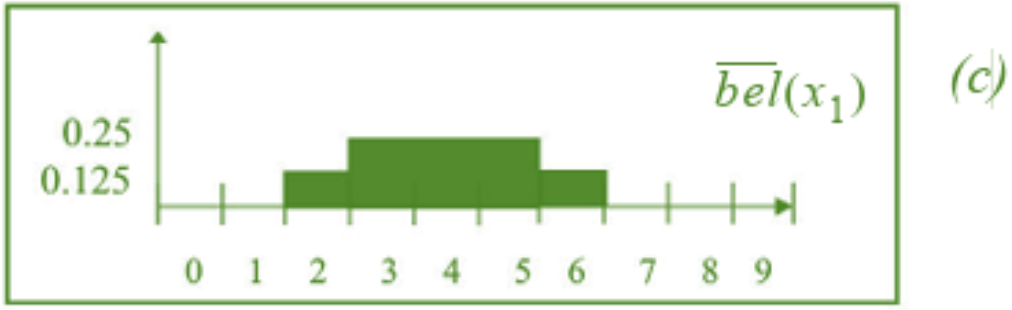
1D case: Prediction update

- The final belief $\overline{bel}(x_1)$ can be computed by using the theorem of total probability.
  - $\overline{bel}(x_1) = \sum_{x_0=0}^{3} p(x_1|u_1, x_0) bel(x_0)$
  - In order to compute the probability of position x_1 in the new belief state, one must sum over all the possible ways in which the robot may reach x_1 according to the potential positions expressed in the former belief state x_0 and the potential input expressed by u_1.
  - Observe that because x_0 is limited between 0 and 3, the robot can only reach the states between 2 to 6



1D case: Prediction update

- The final belief $\overline{bel}(x_1)$ can be computed by using the theorem of total probability.
  - $\overline{bel}(x_1) = \sum_{x_0=0}^{3} p(x_1|u_1, x_0) bel(x_0)$

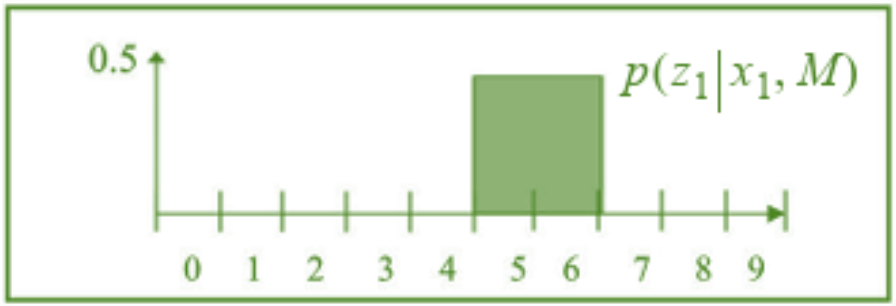


(c)

  - $p(x_1 = 2) = p(x_0 = 0)p(u_1 = 2) = 0.125$
  - $p(x_1 = 3) = p(x_0 = 0)p(u_1 = 3) + p(x_0 = 1)p(u_1 = 2) = 0.25$
  - $p(x_1 = 4) = p(x_0 = 1)p(u_1 = 3) + p(x_0 = 2)p(u_1 = 2) = 0.25$
  - $p(x_1 = 5) = p(x_0 = 2)p(u_1 = 3) + p(x_0 = 3)p(u_1 = 2) = 0.25$
  - $p(x_1 = 6) = p(x_0 = 3)p(u_1 = 3) = 0.125$

1D case: Measurement update

- The robot uses its onboard rangefinder and measures the distance z from the origin.
  - The statistical error model of the range sensor is given as: the distance of the robot from the origin can be equally 5 or 6 units.
  - The final belief is computed accordingly to the Bayes rule.
  - $bel(x_1) = \eta p(z_1|x_1, M)\overline{bel}(x_1)$

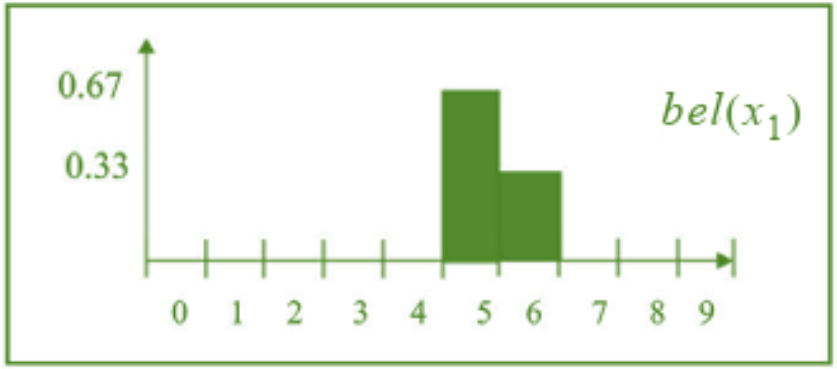


(d)

1D case: Measurement update

- The robot uses its onboard rangefinder and measures the distance z from the origin.
  - $bel(x_1) = \eta p(z_1|x_1, M)\overline{bel}(x_1)$
  - $bel(x_1 = 5) = \eta p(z_1 = 5|x_1 = 5, M)\overline{bel}(x_1 = 5)$
  - $bel(x_1 = 6) = \eta p(z_1 = 6|x_1 = 6, M)\overline{bel}(x_1 = 6)$
  - $\eta = ?$

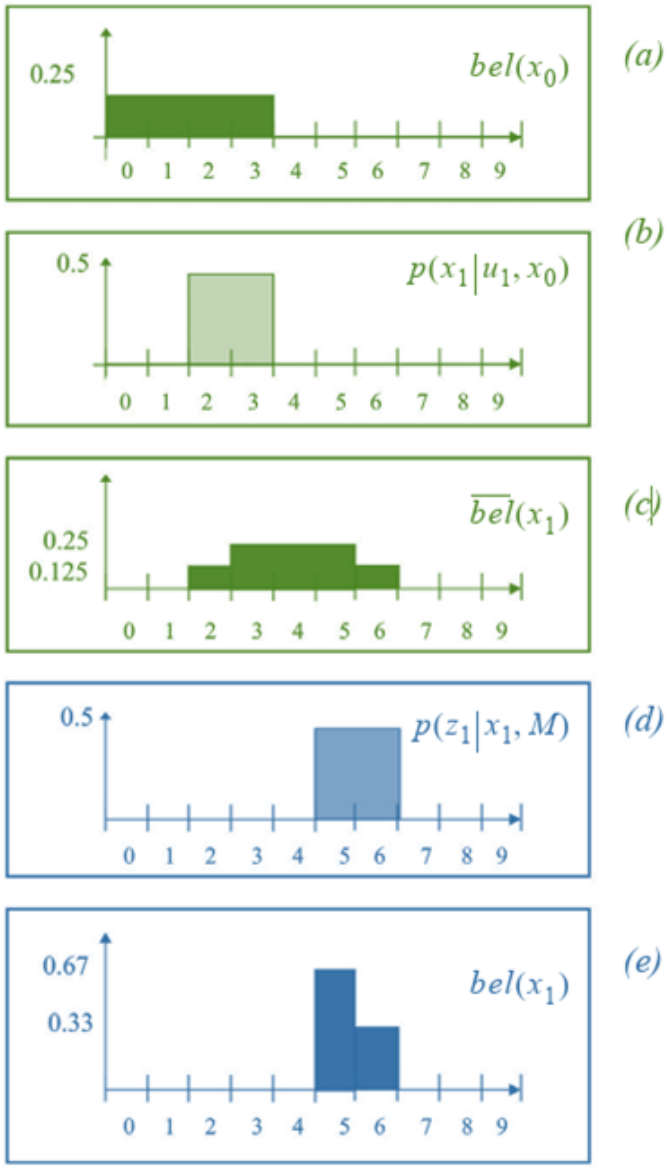




(e)

1D case: Summary

- Two steps procedure.
  - Update prediction:
    $\overline{bel}(x_1)$
    $= \sum_{x_0=0}^{3} p(x_1|u_1, x_0) bel(x_0)$
  - Update measurement:
    $bel(x_1)$
    $= \eta p(z_1|x_1, M) \overline{bel}(x_1)$

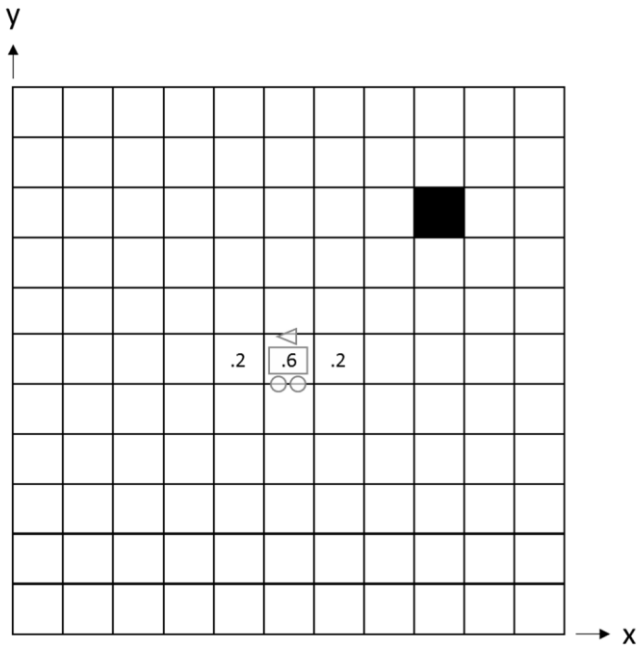


Markov localization: 2D case

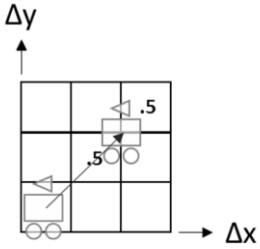
- The real world is at least 2D (moving in the plane)
  - Markov Localization scales badly with the size of the environment
  - Space: 10m x 10m with a grid size of 0.1m and an angular resolution of 1deg
  - → $100 \cdot 100 \cdot 360 = 3.6 \times 10^6$ grid points (states)
  - → prediction step requires in worst case $\left(3.6 \times 10^6\right)^2$ multiplications and summations
- Fine fixed decomposition grids result in a huge state space
  - Very important processing power needed
  - Large memory requirement

Markov localization: 2D case

- Extension to 2D
  - $\overline{bel}(x_t) = \sum_{x_{t-1}} p(x_t|u_t, x_{t-1}) bel(x_{t-1})$
  - $bel(x_t) = \eta p(z_t|x_t, M)\overline{bel}(x_t)$



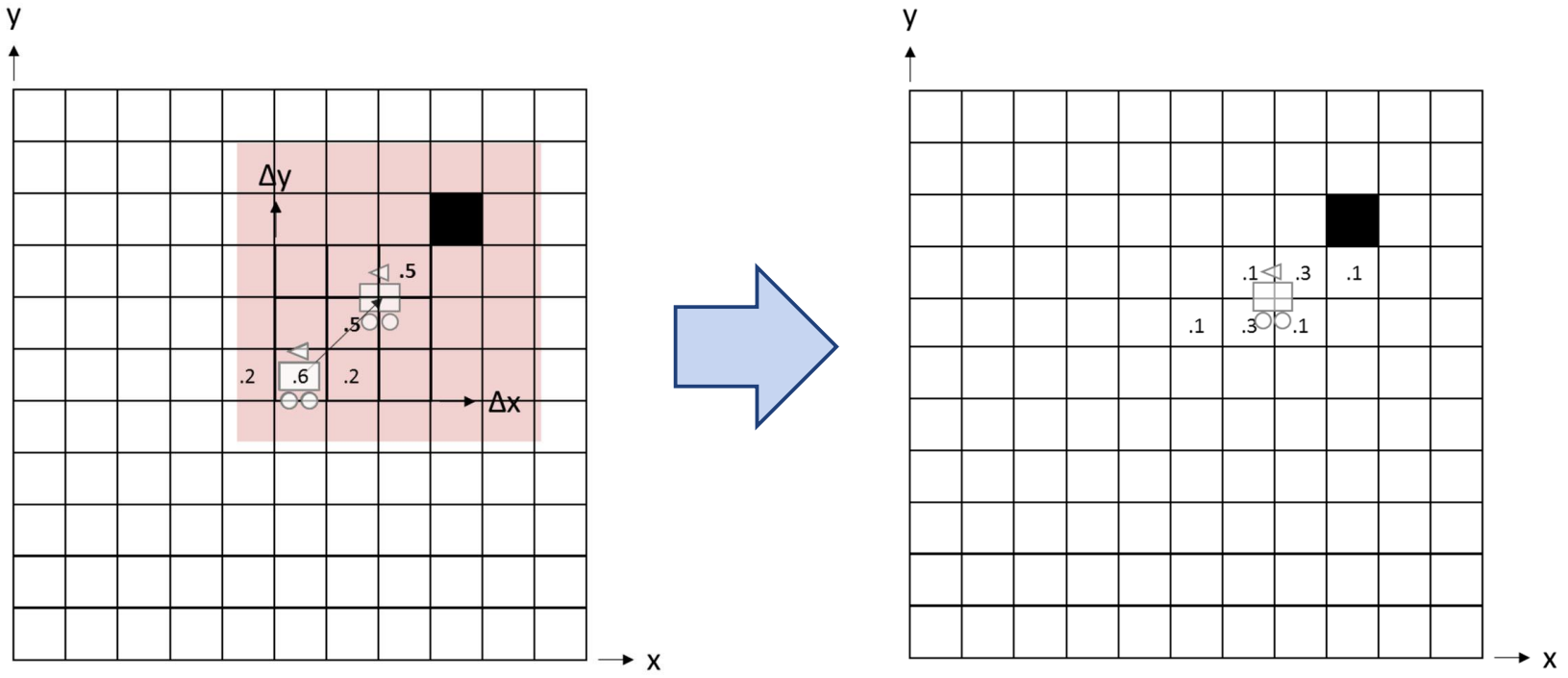
Initial robot's position $bel(x_{t-1})$



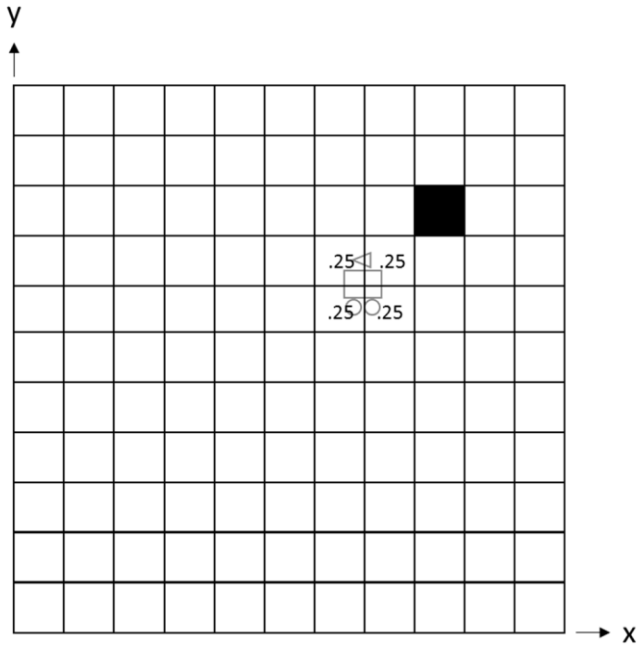
Motion uncertainty $p(x_t|u_t, x_{t-1})$

Markov localization: 2D case

$$\overline{bel}(x_t) = \sum_{x_{t-1}} p(x_t|u_t, x_{t-1}) bel(x_{t-1})$$



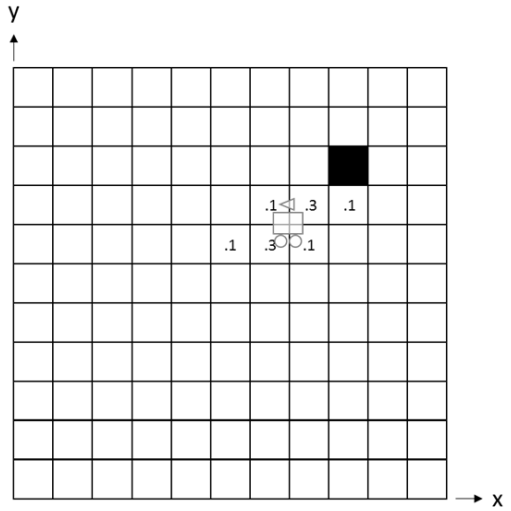
Markov localization: 2D case



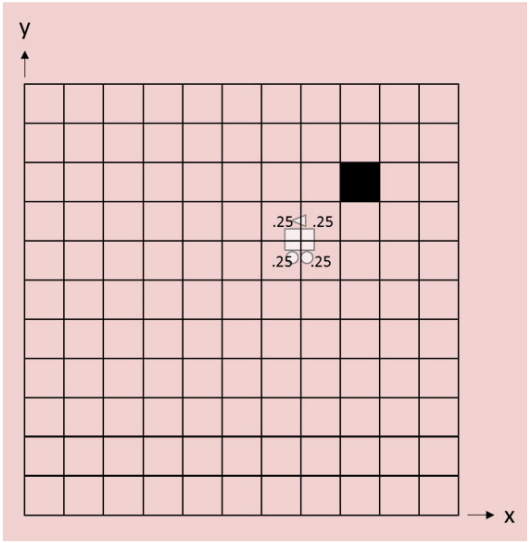
Robot's belief based on measurement $p(z_t|x_t, M)$

Markov localization: 2D case

$$bel(x_t) = \eta p(z_t|x_t, M)\overline{bel}(x_t)$$



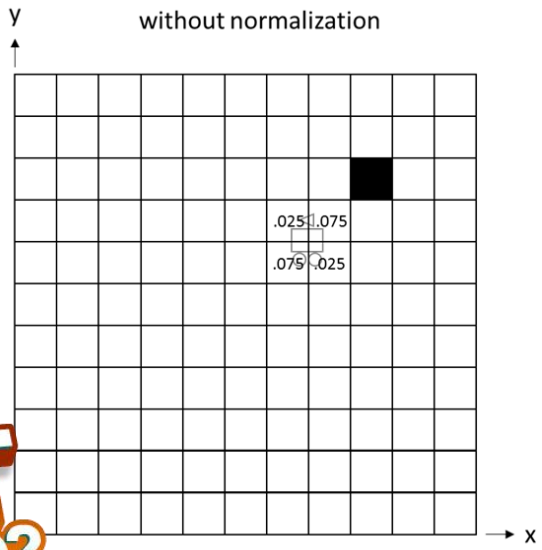
Belief after movement $\overline{bel}(x_t)$

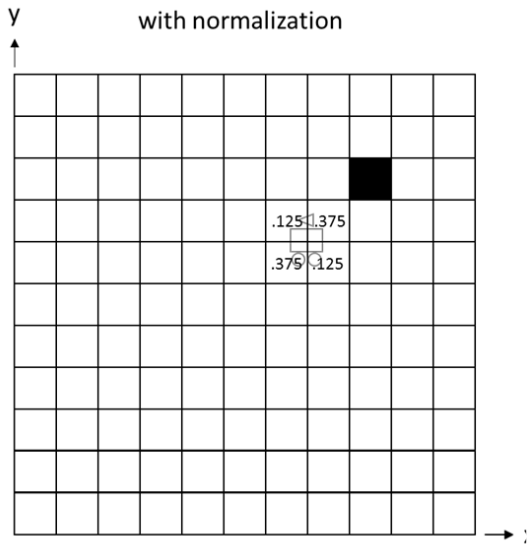


Belief based on measurement $p(z_t|x_t, M)$

Markov localization: 2D case

$$bel(x_t) = \eta p(z_t|x_t, M)\overline{bel}(x_t)$$







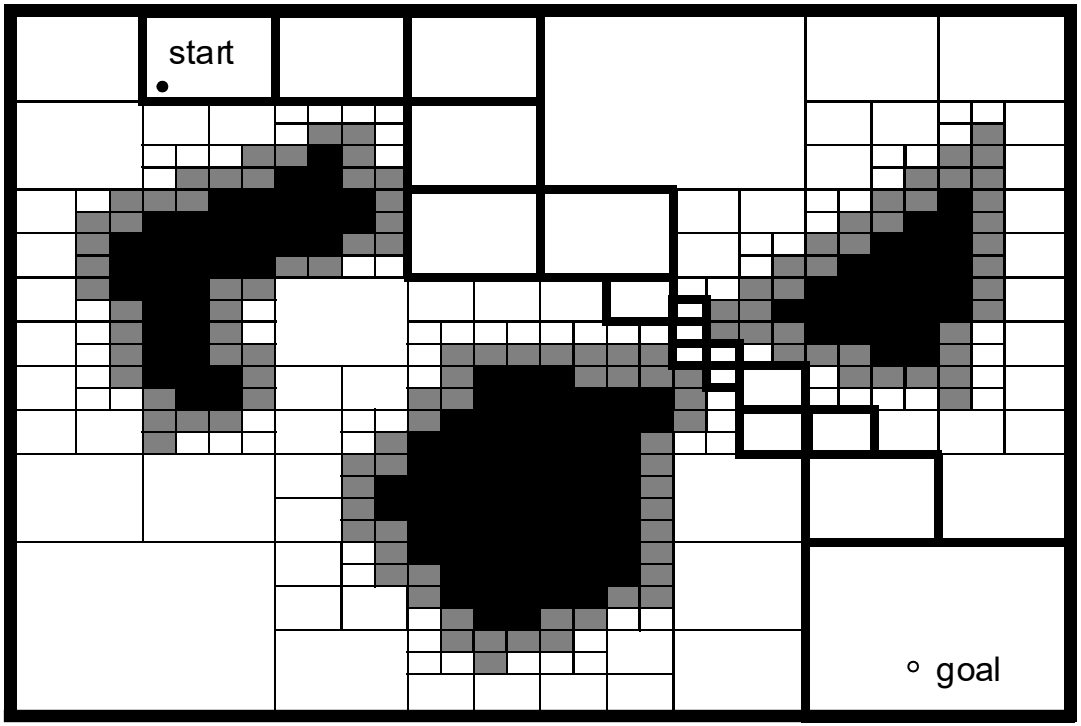
$\eta = p(y)^{-1}$, so that $\sum p = 1$

What is η?

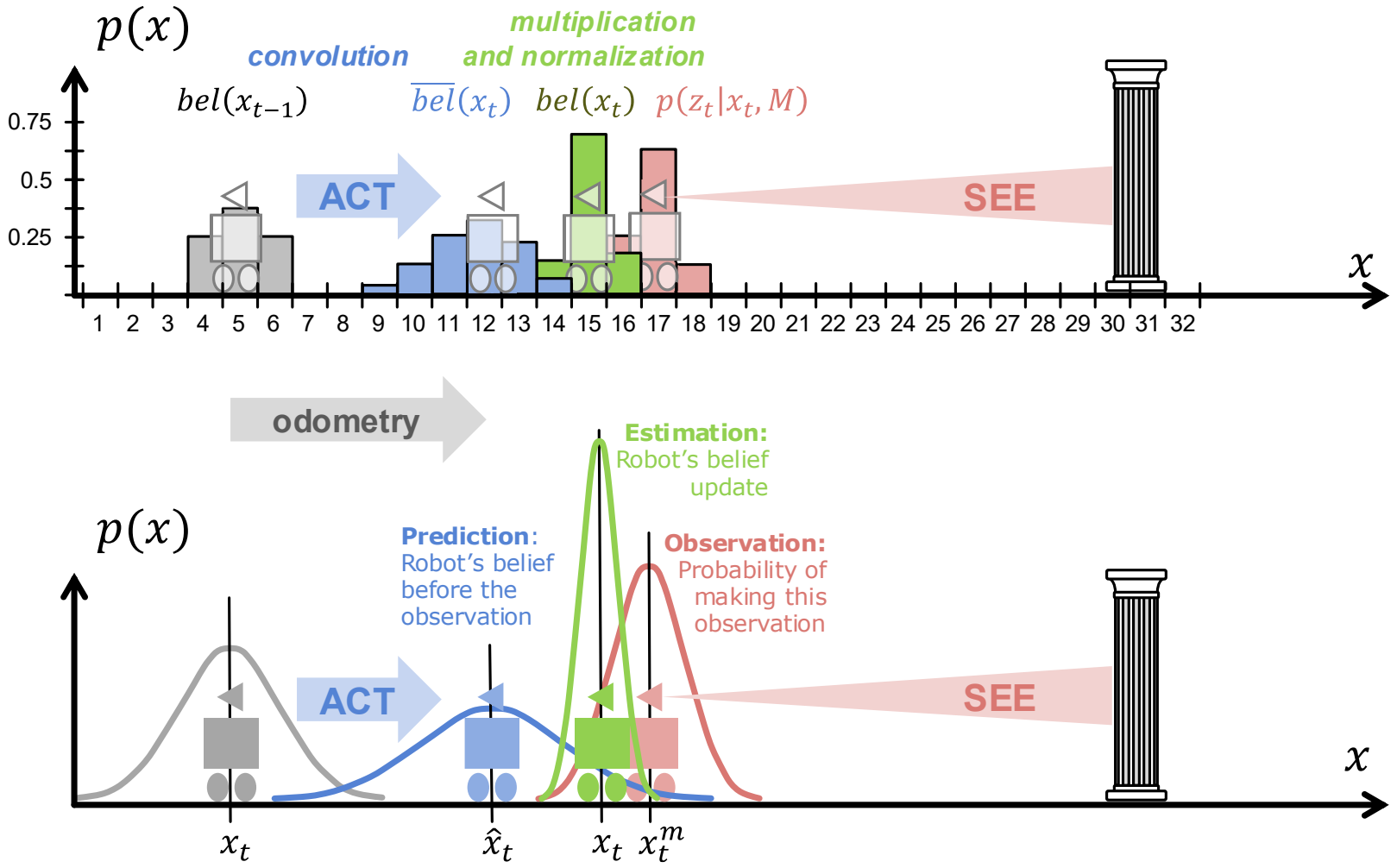


Markov localization: 2D case

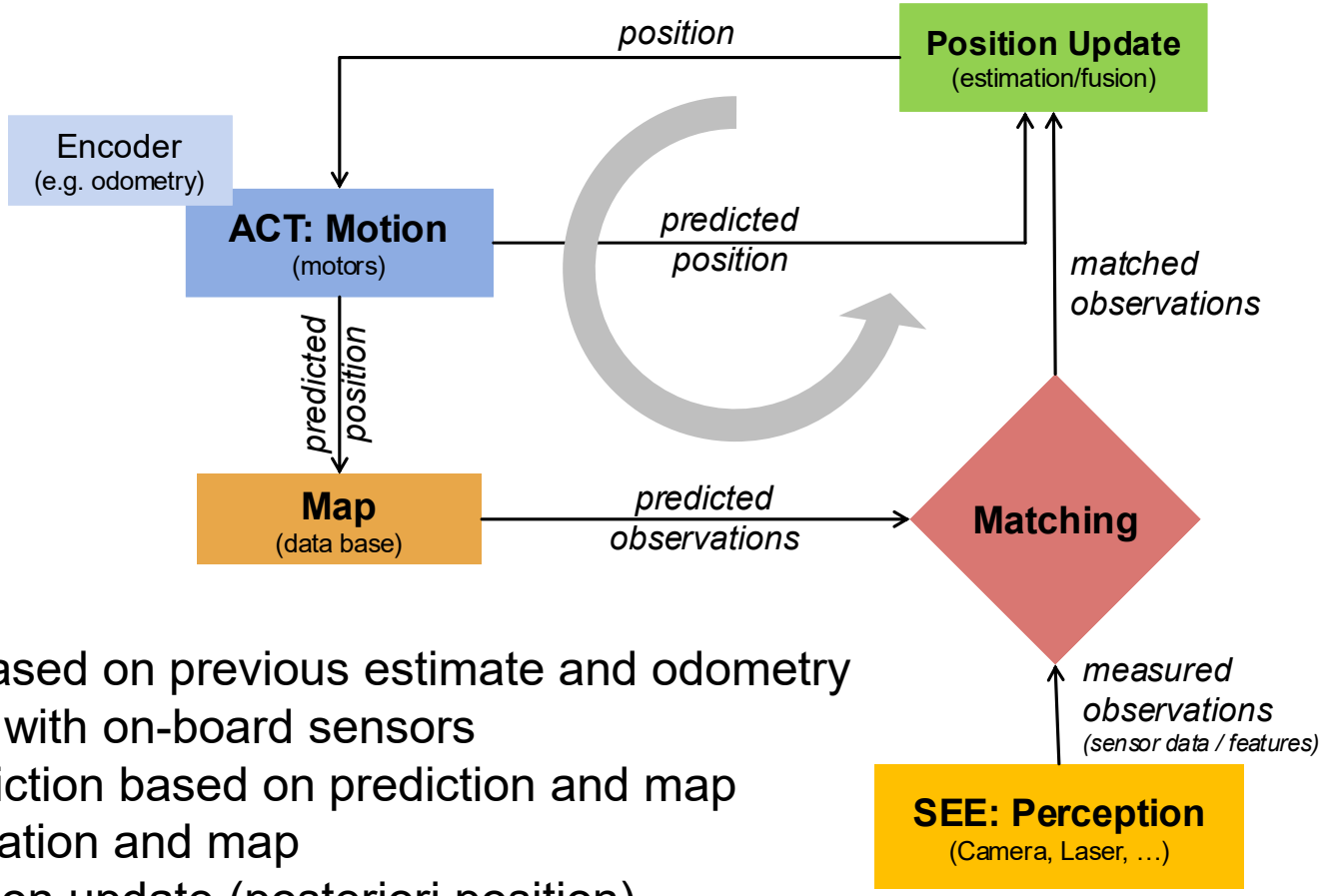
- Reducing computational complexity
  - Adaptive cell decomposition
  - Motion model (Odomety) limited to a small number of grid points
  - Randomized sampling



ACT & SEE | Markov vs Kalman



Kalman filter localization



1. Prediction (ACT) based on previous estimate and odometry
2. Observation (SEE) with on-board sensors
3. Measurement prediction based on prediction and map
4. Matching of observation and map
5. Estimation → position update (posteriori position)

Kalman filter localization

- Continuous pose representation
  - Markov Localization scales badly with the size of the environment
- Kalman Filter Assumption
  - Error approximation with **Gaussian (normal) distribution**
  - Output distribution is a linear function of the input distribution
- Kalman filter localization tracks the robot's belief state typically as a single hypothesis with **normal distribution**
- Kalman filter localization thus addresses the position tracking problem, but not the global localization or the kidnapped robot problem.

Kalman filter localization

- **Goal**: Fuse motion (process) and sensor (measurement) information to estimate the robot state with quantified uncertainty.

- Linear-Gaussian Kalman filter localization model:

  - **State**: $x_k = \mathrm{A}x_{k-1} + \mathrm{B}u_k + \omega_{k-1},\ \omega_{k-1} \sim \mathcal{N}(0, \mathrm{Q})$
  - **Measurement**: $z_k = \mathrm{H}x_{k-1} + v_{k-1},\ v_{k-1} \sim \mathcal{N}(0, \mathrm{R})$
  - Independence: ω and v are zero-mean, mutually independent, and independent of past states.
  - The filter maintains a Gaussian belief $\mathcal{N}(\hat{x}_{k|k}, P_{k|k})$

Prediction update

- Given $\hat{x}_{k-1|k-1}$ and $P_{k-1|k-1}$, incorporate control u_k:
  - **State prediction**: $\hat{x}_{k|k-1} = \mathrm{A}\hat{x}_{k-1|k-1} + \mathrm{B}u_k$,
  - **Measurement**: $P_{k|k-1} = AP_{k-1|k-1}A^T + Q$,
  - Independence: ω and v are zero-mean, mutually independent, and independent of past states.
  - Intuition: uncertainty grows by process noise Q and model propagation.

Measurement correction update

- Given measurement z_k :
  - One can obtain **the innovation (residual)**: $\tilde{y}_k = z_k - \mathrm{H}\,\hat{x}_{k|k-1}$,
  - **Innovation covariance**: $S_k = H\,P_{k|k-1}H^T + R$,
  - Kalman gain is defined as: $K_k = P_{k|k-1}H^T\,S_k^{-1}$,
  - State update: $\hat{x}_{k|k} = \mathrm{A}\hat{x}_{k|k-1} + K_k\tilde{y}_k$,
  - Covariance update: $P_{k|k} = (I - K_k H)\,P_{k|k-1}$
- We can observe that K_k balances trust between the prediction and the measurement via "Q" vs. "R".

Algorithm (summary)

- **Predict**:

$$\hat{x}_{k|k-1} = \mathrm{A}\hat{x}_{k-1|k-1} + \mathrm{B}u_k, \text{ and } P_{k|k-1} = AP_{k-1|k-1}A^T + Q$$

- **Compute innovation**:

$$\tilde{y}_k = z_k - \mathrm{H}\,\hat{x}_{k|k-1}, \text{ and } S_k = H\,P_{k|k-1}H^T + R$$

- **KL Gain**:

$$K_k = P_{k|k-1}H^T\,S_k^{-1}$$

- **Correct**:

$$\hat{x}_{k|k} = \mathrm{A}\hat{x}_{k|k-1} + K_k\tilde{y}_k, \text{ and } P_{k|k} = (I - K_kH)\,P_{k|k-1}$$

A minimal 1-D Kalman Filter

- In one dimensional space, A, B, H, Q, R are all scalars
- Typical 1-D specification:

$$A=1,\ B=1,\ H=1,\ Q>0,\ R>0.$$

- Scalar equations become:
  - **Predict**: $\hat{x}_{k|k-1}= \hat{x}_{k-1|k-1}+ u_k$, and $P_{k|k-1}= P_{k-1|k-1}+Q$
  - **Updates**: $\tilde{y}_k= z_k - \hat{x}_{k|k-1}$, and $K_k= \frac{P_{k|k-1}}{P_{k|k-1}+R}$

$$\hat{x}_{k|k}= \hat{x}_{k|k-1}+K_k\tilde{y}_k, \text{ and } P_{k|k}= (1 - K_k)\, P_{k|k-1}$$

- All quantities are scalars, so transposes/inverses are trivial.

Example: 1-D Kalman kilter

- For an one dimensional system, given $\mathrm{A} = \mathrm{B} = \mathrm{H} = 1$, $\mathrm{Q} = 0.05, \mathrm{R} = 0.2$
- Assume that $\hat{x}_0 = 0, P_0 = 1$, controls $u_1 = u_2 = 1$, and measurements $z_1 = 0.9, z_2 = 2.2$.
- Please try to show the $\hat{x}_{k|k}$ and $P_{k|k}$ for $k = 1,2$





Example: 1-D Kalman kilter

- $k = 1$, based on the given values of parameters
  - **Predict**:

$\hat{x}_{1|0} = \hat{x}_0 + u_1 = 0 + 1 = 1$, and $P_{1|0} = P_0 + Q = 1 + 0.05 = 1.05$

  - **Updates**:

$\tilde{y}_1 = z_1 - \hat{x}_{1|0} = 0.9 - 1 = -0.1$, and $K_1 = \frac{P_{1|0}}{P_{1|0} + R} = \frac{1.05}{1.05 + 0.2} = 0.84$

$\hat{x}_{1|1} = \hat{x}_{1|0} + K_1 \tilde{y}_1 = 1 + 0.84 \times (-0.1) = 0.916$, and

$P_{1|1} = (1 - K_1)\, P_{1|0} = = (1 - 0.84) \times 1.05 = 0.168$

Example: 1-D Kalman kilter

- $k = 2$, based on the given values of parameters
  - **Predict**:

$\hat{x}_{2|1} = \hat{x}_{1|1} + u_2 = 0.916 + 1 = 1.916$, and $P_{2|1} = P_{1|1} + Q = 0.168 + 0.05 = 0.218$

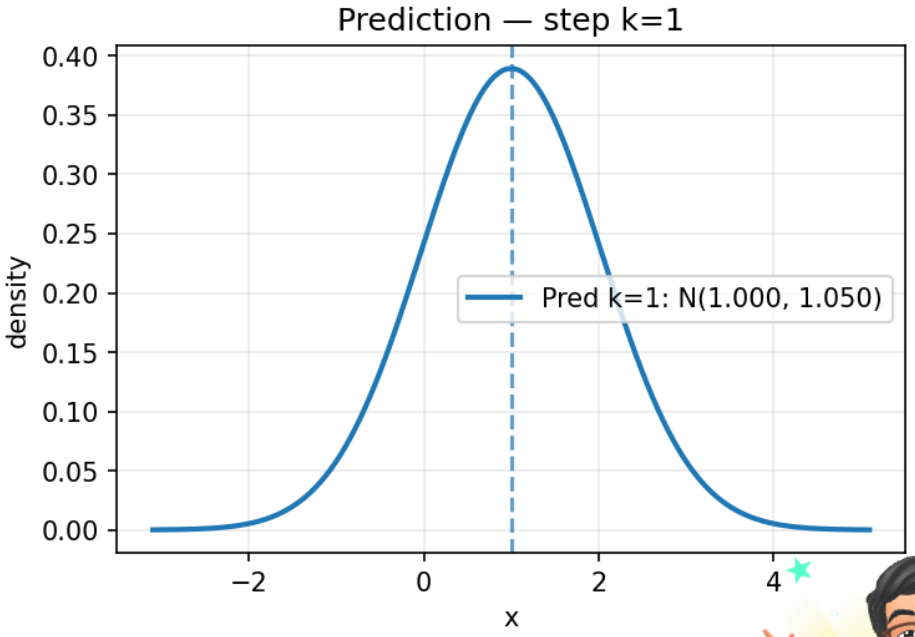
  - **Updates**:

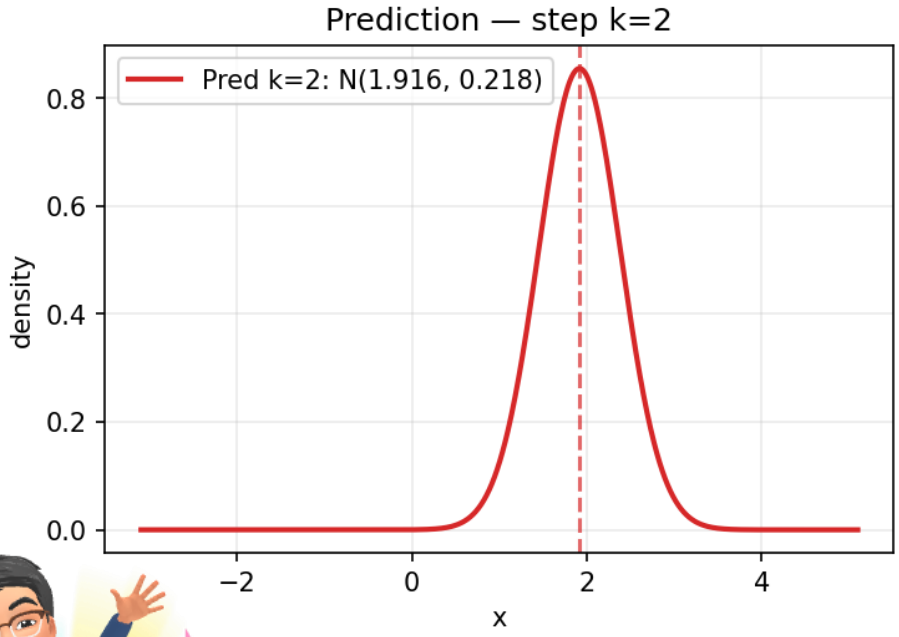
$\tilde{y}_2 = z_2 - \hat{x}_{2|1} = 2.2 - 1.916 = 0.284$, and $K_2 = \frac{P_{2|1}}{P_{2|1} + R} = \frac{0.218}{0.218 + 0.2} \approx 0.52153$

$\hat{x}_{2|2} = \hat{x}_{2|1} + K_2 \tilde{y}_2 = 1.916 + 0.52153 \times 0.284 \approx 2.0641$, and

$P_{2|2} = (1 - K_2)\, P_{2|1} = (1 - 0.52153) \times 0.218 \approx 0.1043$

Example: 1-D Kalman kilter





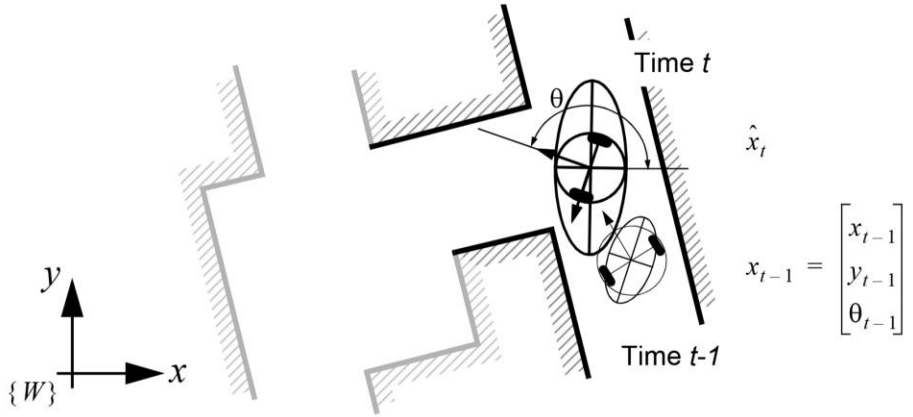


Kalman Filter Localization

- Prediction Update
- Robot's position prediction:

$$\hat{x}_t = \mathrm{f}(x_{t-1}, u_t) = \begin{bmatrix} x_{t-1} \\ y_{t-1} \\ \theta_{t-1} \end{bmatrix} + \begin{bmatrix} \frac{\Delta s_r + \Delta s_l}{2} cos\left(\theta + \frac{\Delta s_r + \Delta s_l}{2b}\right) \\ \frac{\Delta s_r + \Delta s_l}{2} cos\left(\theta + \frac{\Delta s_r + \Delta s_l}{2b}\right) \\ \frac{\Delta s_r + \Delta s_l}{b} \end{bmatrix}$$

  - The robot's position $\hat{x}_t$ is predicted based on its old location x_{t-1} and its movement u_t



Kalman Filter Localization

- Updated covariance matrix:

$$\widehat{P}_t = F_x \cdot P_{t-1} \cdot F_x^T + F_u \cdot Q_t \cdot F_u^T$$

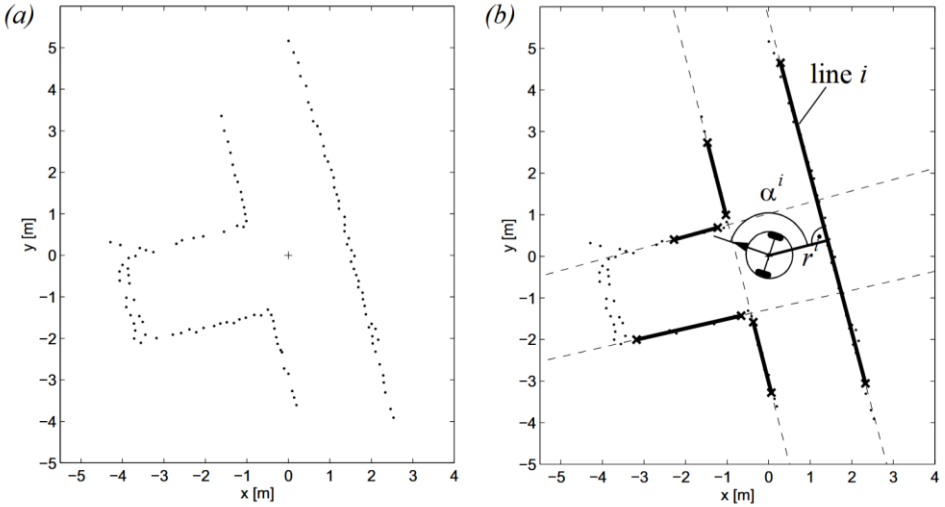
  - F and F represent the Jacobians of the function f with respect to x_t and u_t
  - Q_t is the covariance of the noise, that is $Q_t = \begin{bmatrix} k_r|\Delta s_r| & 0 \\ 0 & k_l|\Delta s_l| \end{bmatrix}$

- $z_t^i = \begin{bmatrix} \alpha_t^i \\ r_t^i \end{bmatrix}$
  - each single observation (i.e., a line feature) is extracted from the raw laser rangefinder data and consists of the two line parameters α_t^i and r_t^i

- $R_t^i = \begin{bmatrix} \sigma_{\alpha\alpha} & \sigma_{\alpha r} \\ \sigma_{r\alpha} & \sigma_{rr} \end{bmatrix}^i$
  - a covariance matrix for each line that can be calculated from the uncertainties of all the measurement points contributing to each line

Kalman Filter Localization

- **Observation**: From the raw data (a) acquired by the laser scanner at time t, lines are extracted (b). The line parameters α^i and r^i and its uncertainties can be represented in the model space (c).



Kalman Filter Localization

- Measurement prediction

$$\hat{z}_t^j = \begin{bmatrix} \hat{\alpha}_t^j \\ \hat{r}_t^j \end{bmatrix} = h^j(\hat{x}_t, m^j) = \begin{bmatrix} {}^{\{W\}}\alpha_t^j - \hat{\theta}_t \\ {}^{\{W\}}r_t^j - (\hat{x}_t \cos({}^{\{W\}}\alpha_t^j) + \hat{y}_t \sin({}^{\{W\}}\alpha_t^j)) \end{bmatrix}$$

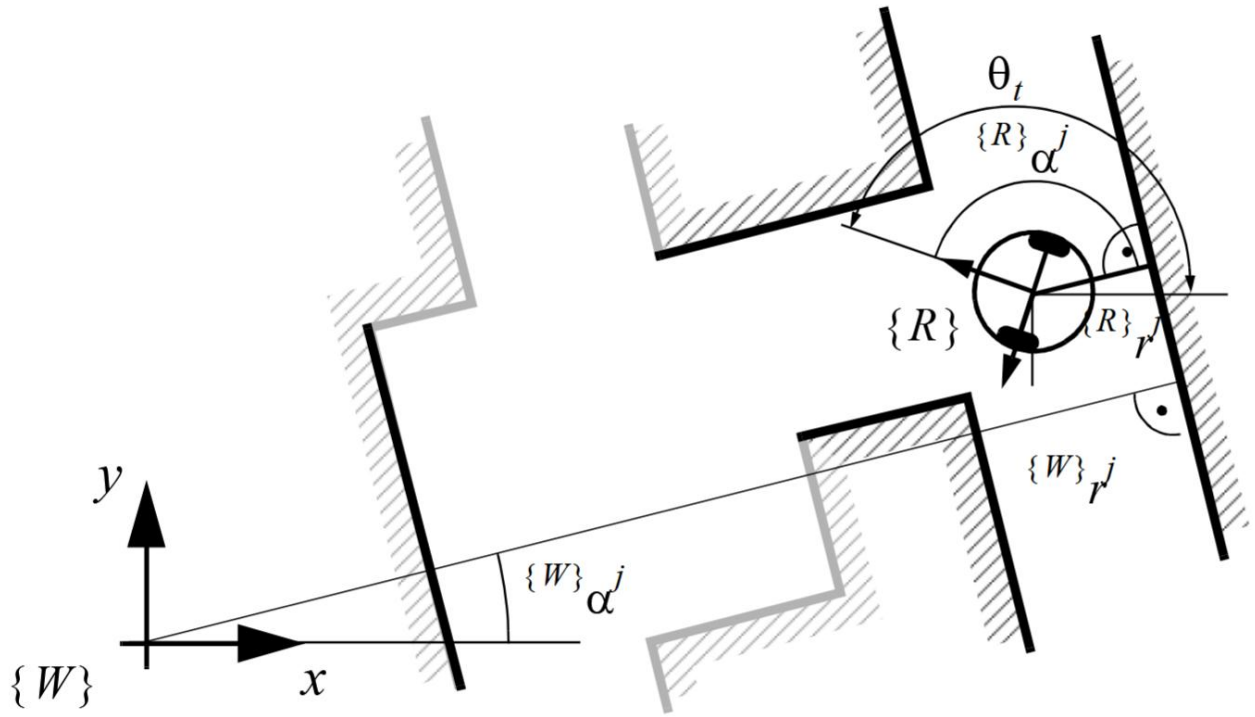
  - transform all the line features in the map into its local robot coordinate frame

$$H^j = \begin{bmatrix} \frac{\partial \alpha_t^j}{\partial \hat{x}} & \frac{\partial \alpha_t^j}{\partial \hat{y}} & \frac{\partial \alpha_t^j}{\partial \hat{\theta}} \\ \frac{\partial r_t^j}{\partial \hat{x}} & \frac{\partial r_t^j}{\partial \hat{y}} & \frac{\partial r_t^j}{\partial \hat{\theta}} \end{bmatrix} = \begin{bmatrix} 0 & 0 & -1 \\ -\cos({}^{\{W\}}\alpha_t^j) & -\sin({}^{\{W\}}\alpha_t^j) & 0 \end{bmatrix}$$

  - Jacobian H^i

Kalman Filter Localization

- Representation of the target position in the world coordinate frame and robot coordinate frame.



$$\hat{x}_t = \begin{bmatrix} \hat{x}_t \\ \hat{y}_t \\ \hat{\theta}_t \end{bmatrix}$$

Kalman Filter Localization

- Matching

$$v_t^{ijT} \cdot (\Sigma_{IN_t}^{ij})^{-1} \cdot v_t^{ij} \leq g^2$$

with

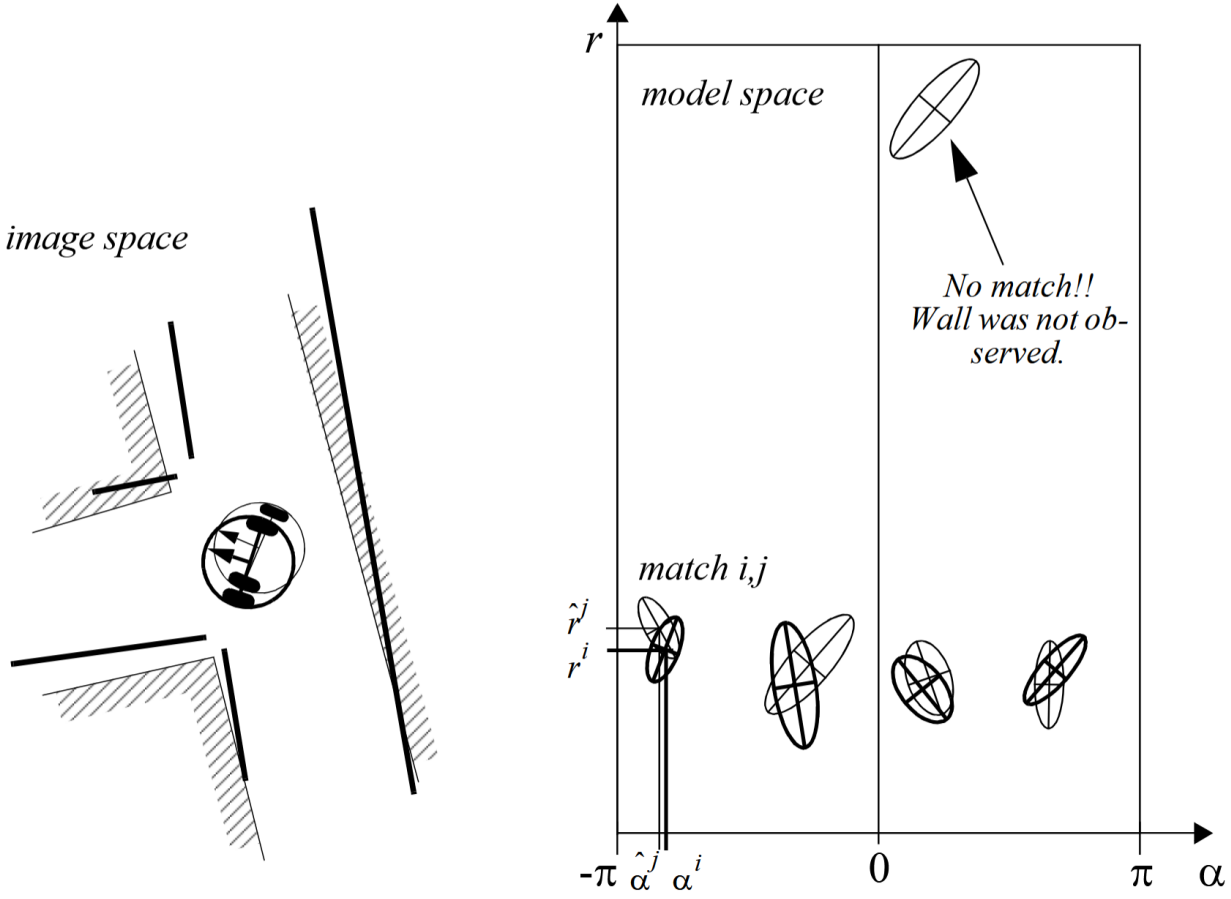
$$v_t^{ij} = [z_t^i - \hat{z}_t^j] = [z_t^i - h^j(\hat{x}_t, m^j)] =$$

$$= \begin{bmatrix} \alpha_t^i \\ r_t^i \end{bmatrix} - \begin{bmatrix} {}^{\{W\}}\alpha_t^j - \hat{\theta}_t \\ {}^{\{W\}}r_t^j - (\hat{x}_t \cos({}^{\{W\}}\alpha_t^j) + \hat{y}_t \sin({}^{\{W\}}\alpha_t^j)) \end{bmatrix},$$

$$\Sigma_{IN_t}^{ij} = H^j \cdot \hat{P}_t \cdot H^{jT} + R_t^i$$

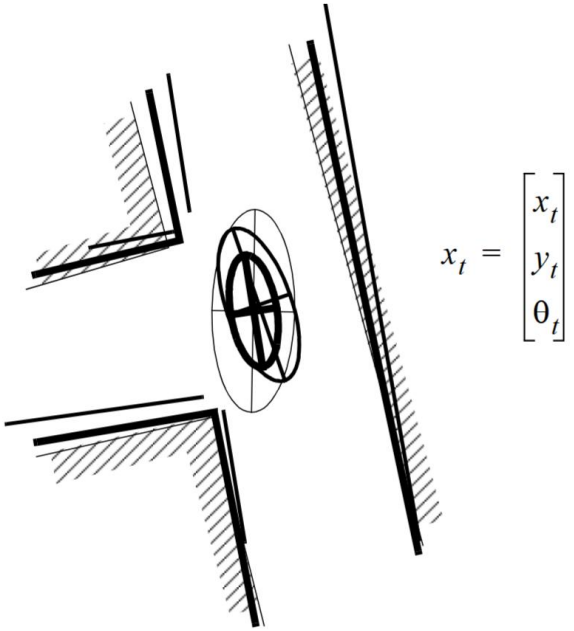
Kalman Filter Localization

- **Matching**: The observations (thick) and the predicted observation (thin) are matched and the innovation and its uncertainties are calculated.



Kalman Filter Localization

- Estimation
- Applying the Kalman filter results in a final pose estimate corresponding to the weighted sum of
  - the pose estimates of each matched pairing of observed and predicted features;
  - the robot position estimation based on odometry and observation positions.



ROSE5730 Autonomous Mobile Robots

Lecture 5: Localization



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